EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Introduction of GOC and Intermediates

1. CH_3CH_2Cl undergoes homolytic fission to produce:-

(A) $CH_3\dot{C}H_2$ and $\dot{Cl}$ (B) $CH_3\dot{C}H_2$ and $Cl^{\ominus}$

(C) $CH_3\overset{\oplus}{C}H_2$ and $Cl^{\ominus}$ (D) $CH_3\overset{\ominus}{C}H_2$ and $Cl^{\oplus}$

CED097

2. Which of the following intermediate has complete octet :-

(A) Carbocation (B) Carbanion (C) Free radical (D) Carbene

CED098

3. In the compound, $\overset{I}{CH_3}-\overset{II}{CH}=\overset{III}{CH}-\overset{IV}{C}\equiv N$, the most electronegative carbon is :

(A) I (B) II (C) III (D) IV

CED132

Inductive effect (Groups showing +I and -I effects)

4. Which statement is correct for inductive effect :

(A) It is a permanent effect

(B) It is the property of single bond

(C) It causes permanent polarisation in the molecule

(D) All are correct

Rankers Academy JEE

CED091

5. Inductive effect involves

(A) displacement of σ–electron (B) delocalisation of π–electron

(C) delocalisation of σ–electrons (D) displacement of π–electrons

CED092

6. Most stable carbocation is

(A) $CH_3-\overset{\oplus}{C}H_2$ (B) $\overset{\oplus}{C}H_2CHCl_2$ (C) $Cl-CH_2-\overset{\oplus}{C}H_2$ (D) $O_2N-CH_2-\overset{\oplus}{C}H_2$

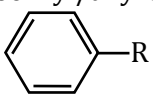
CED093

7. A carbocation in which no dispersal of charge occurs is

(A) $R_3C^{\oplus}$ (B) $R_2CH^{\oplus}$ (C) $RCH_2^{\oplus}$ (D) $^{\oplus}CH_3$

CED094

8. In the following benzyl/allyl system

$R-CH=CH_2$ or 

(R is alkyl group)

decreasing order of inductive effect is :

(A) $(CH_3)_3C- > (CH_3)_2CH- > CH_3CH_2-$ (B) $CH_3CH_2- > (CH_3)_2CH- > (CH_3)_3C-$

(C) $(CH_3)_2CH- > CH_3CH_2- > (CH_3)_3C-$ (D) $(CH_3)_2C- > CH_3CH_2- > (CH_3)_3CH-$

CED095

For More Join: @IITJEE_Advanced

9. Most stable carbocation is :

(A) $CH_3-\overset{\oplus}{C}H_2$ (B) $Br-CH_2-\overset{\oplus}{C}H_2$ (C) $Cl-CH_2-\overset{\oplus}{C}H_2$ (D) $F-CH_2-\overset{\oplus}{C}H_2$

CED099

10. Most stable carbanion is :

(A) $\overset{\ominus}{C}H_2-CH_2-NO_2$ (B) $\overset{\ominus}{C}H_2-CH_2-CN$

(C) $\overset{\ominus}{C}H_2-CH_2-COOH$ (D) $\overset{\ominus}{C}H_2-CH_2-CH_2-CH_2-F$

CED100

Resonance, Conditions for resonance, Stability of Resonating structures

11. If A is $Ph\overset{\ominus}{C}H_2$ and B is $CH_2=CH-\overset{\ominus}{C}H_2$, the greater number of resonating structures is of –

(A) A (B) B (C) both A and B (D) None of these

CED116

12. Which of the following pairs represent resonance?

(A) $CH_2=CHOH$; CH_3CHO (B) $\overset{\ominus}{C}H_2-CHO$; $H_2C=CH-\overset{\ominus}{O}$

(C) $CH_3-\overset{O}{\overset{\|}{C}}-CH_3$, $CH_3-\overset{OH}{\overset{|}{C}}=CH_2$ (D) $CH_3-\overset{\oplus OH}{\overset{\|}{C}}-CH_3$, $CH_3-\overset{OH}{\overset{|}{\underset{H}{\underset{|}{C}}}}-CH_3$

CED117

13. Which of the following groups cannot participate in resonance with other suitable group :

(A) – COOH (B) –$COOCH_3$ (C) –COCl (D) $-\overset{\oplus}{N}H_3$

CED119

14. Which one of the following pair of structures does not represent the phenomenon of resonance?

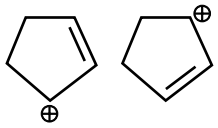
(A) $H_2C=CH-\overset{O}{\overset{\|}{C}}-H$; $\overset{\oplus}{C}H_2-CH=\overset{O^{\ominus}}{\overset{|}{C}}-H$ (B) $CH_2=CH-\overset{\oplus}{C}HCl$; $\overset{\oplus}{C}H_2-CH=CH-Cl$

(C) $CH_3-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{O}$; $CH_3-\overset{O^{\ominus}}{\overset{|}{C}}=O$ (D) $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-CH_3$; $CH_3-CH=\overset{HO}{\overset{|}{C}}-CH_3$

CED123

15. Which of the following pairs of structure is not a pair of resonating structure?

(A) $H-C\equiv\overset{\oplus}{N}H$; $H-\overset{\oplus}{C}=\ddot{N}H$ (B) $CH_3CH=CHCH_3$; $CH_3CH_2CH=CH_2$

(C) $CH_3-\underset{O^{\ominus}}{\underset{|}{C}}=CH-\overset{O}{\overset{\|}{C}}-CH_3$; $CH_3-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{C}H-\overset{O}{\overset{\|}{C}}-CH_3$ (D) 

CED127

16. The decreasing order of stability of the following resonating structures

$CH_2=CH-\underset{(I)}{\ddot{\underset{\cdot\cdot}{Cl}}:}$ $\longleftrightarrow$ $:\overset{\ominus}{C}H_2-CH=\underset{(II)}{\overset{\oplus}{\ddot{\underset{\cdot\cdot}{Cl}}}:}$ $\longleftrightarrow$ $\overset{\oplus}{C}H_2-CH=\underset{(III)}{\ddot{\underset{\cdot\cdot}{\overset{\ominus}{Cl}}}:}$ is :

(A) I > II > III (B) II > III > I (C) III > II > I (D) I > III > II

CED115

Rankers Academy JEE
For More Join: @IITJEE_Advanced

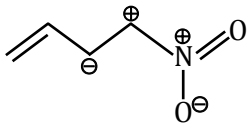
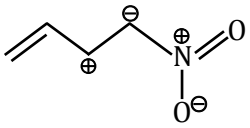
17. Which of the following is least stable :

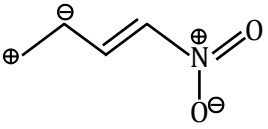
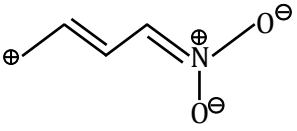
(A) $CH_3-\overset{\oplus}{O}=CH-\overset{\ominus}{C}H-CH=CH_2$ (B) $CH_3-\overset{\oplus}{O}=CH-CH=CH-\overset{\ominus}{C}H_2$

(C) $CH_3-O-\overset{\oplus}{C}H-\overset{\ominus}{C}H-CH=CH_2$ (D) $CH_3-\ddot{\underset{\cdot\cdot}{O}}-\overset{\ominus}{C}H-\overset{\oplus}{C}H-CH=CH_2$

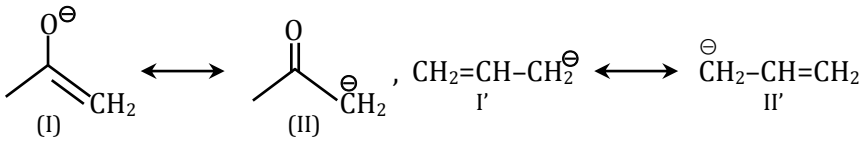
CED121

18. Among the following, the least stable resonance structure is –

(A)  (B) 

(C)  (D) 

CED122

19. 

The correct order of stability -

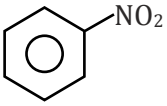
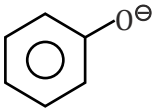
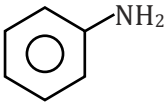
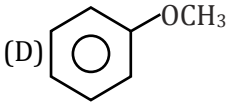
(A) I > II , I' = II' (B) I < II , I' < II' (C) I = II , I' = II' (D) I > II , II' > I'

CED124

Rankers Academy JEE

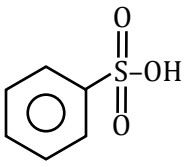
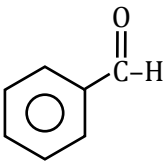
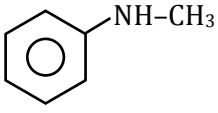
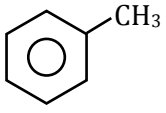
Mesomeric effect (Groups showing +M and -M effects)

20. In which of the following molecules π – electron density in ring is maximum:

(A)  (B)  (C)  (D) 

CED134

21. Which of the following contains group with +M effect?

(A)  (B)  (C)  (D) 

CED135

22. Which of the following shows correct order of –M effect

(A) $-\overset{O}{\overset{\|}{C}}-H > -\overset{O}{\overset{\|}{C}}-CH_3 > -\overset{O}{\overset{\|}{C}}-Cl$ (B) $-\overset{O}{\overset{\|}{C}}-Cl > -\overset{O}{\overset{\|}{C}}-CH_3 > -\overset{O}{\overset{\|}{C}}-H$

(C) $-\overset{O}{\overset{\|}{C}}-Cl > -\overset{O}{\overset{\|}{C}}-H > -\overset{O}{\overset{\|}{C}}-CH_3$ (D) $-\overset{O}{\overset{\|}{C}}-CH_3 > -\overset{O}{\overset{\|}{C}}-H > -\overset{O}{\overset{\|}{C}}-Cl$

CED136

For More Join: @IITJEE_Advanced

23. Match the column :

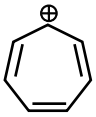
	Column-I		**Column-II**
(A)	Group donate $e^{\ominus}$ inductively but does not donate / withdraw by resonance	(P)	–OH
(B)	Group withdraw $e^{\ominus}$ inductively but does not donate / withdraw by resonance	(Q)	$-NO_2$
(C)	Group withdraw $e^{\ominus}$ inductively & donate $e^{\ominus}$ by resonance	(R)	$-CH_2-CH_3$
(D)	Group withdraw $e^{\ominus}$ inductively & withdraw $e^{\ominus}$ by resonance	(S)	$-\overset{\oplus}{N}H_3$
		(T)	$-NH_2$

(A) (A) → R (B) → S (C) → P, T (D) → Q
(B) (A) → R (B) → Q (C) → R (D) → P,Q
(C) (A) → P (B) → Q (C) → R (D) → S
(D) (A) → P,Q (B) → S (C) → R (D) → Q

CED139

Aromaticity

24. The most stable and the least stable carbocation among

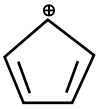
(I)  (II) $CH_2=CH-\overset{\oplus}{C}H_2$ (III) $C_6H_5-\overset{\oplus}{C}H_2$ (IV) $CH_3-\overset{\oplus}{C}H-CH_3$

are respectively :

(A) II, I (B) III, IV (C) I, II (D) I, IV

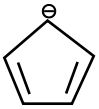
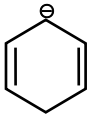
CED149

25. Which of the structure below would be aromatic?

(A)  (B)  (C)  (D) 

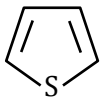
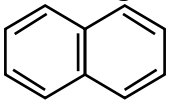
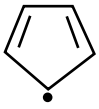
CED152

26. Which of the following carbanion is aromatic?

(A)  (B)  (C)  (D) 

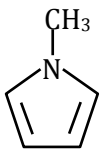
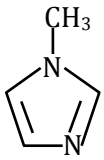
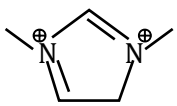
CED153

27. The non–aromatic compound among the following is :

(A)  (B)  (C)  (D) 

CED154

28. Choose the aromatic nitrogen heterocycles.

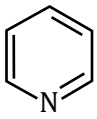
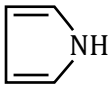
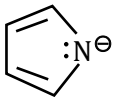
(i)  (ii)  (iii) 

(A) (i) (B)(ii) (C) (iii) (D) (i) and (ii)

CED156

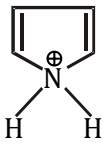
Rankers Academy JEE
For More Join: @IITJEE_Advanced

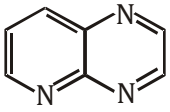
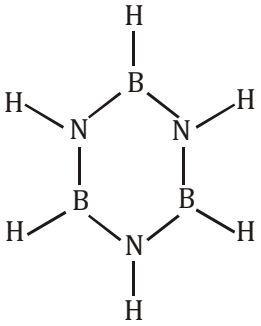
29. Which species is non-aromatic ?

(A)  (B)  (C)  (D) 

CED157

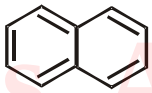
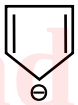
30. Which of the following is non-aromatic :

(A)  (B) 

(C)  (D) 

CED158

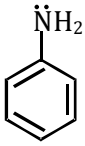
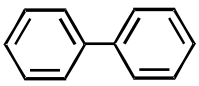
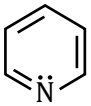
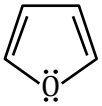
31. Which of the following is nonaromatic

(A)  (B)  (C)  (D) 

Rankers Academy JEE

CED159

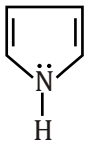
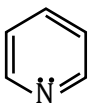
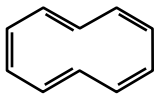
32. Which of the following compounds are aromatic in nature

(a)  (b)  (c)  (d) 

(A) Only a (B) Only a and b (C) Only a, b, d (D) All of these

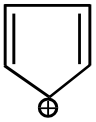
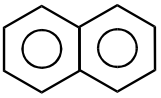
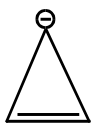
CED160

33. Which of the following is non-aromatic compound

(A)  (B)  (C)  (D) None of these

CED161

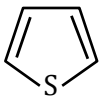
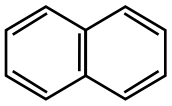
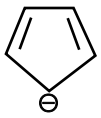
34. Among the following the aromatic compound is –

(A)  (B)  (C)  (D) 

CED162

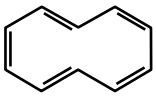
For More Join: @IITJEE_Advanced

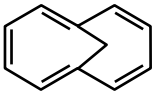
35. The non-aromatic compound among the following is :-

(A)  (B)  (C)  (D) 

CED163

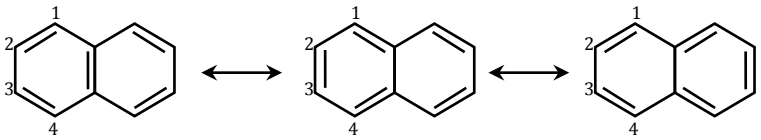
36. For the given compounds, the correct order is :


(I)


(II)

(A) Aromatic, aromatic
(B) non-aromatic, aromatic
(C) anti-aromatic, non-aromatic
(D) Aromatic, non-aromatic

CED164

37. 

These are three canonical structures of naphthalene. Examine them and find correct statement among the following :

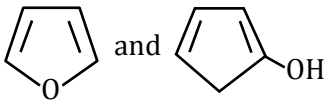
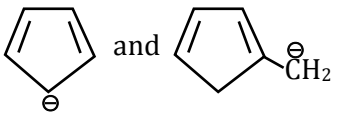
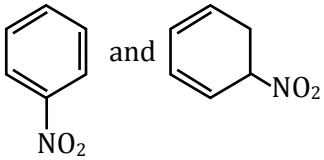
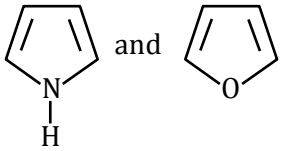
(A) All C–C bonds are of same length
(B) C_1–C_2 bond is shorter than C_2–C_3 bond.
(C) C_1–C_2 bond is longer than C_2–C_3 bond
(D) None

CED169

Rankers Academy JEE

Resonance energy

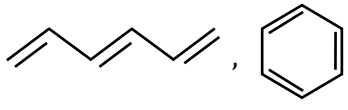
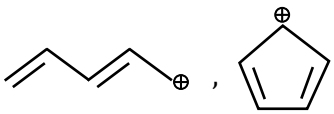
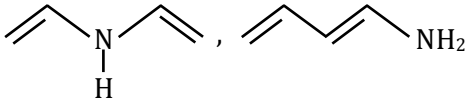
38. In the given pair of compounds select the one in which second compound having lesser resonance energy :

(i) 
(ii) 
(iii) 
(iv) 

(A) i and ii
(B) iii
(C) i, ii and iii
(D) i, ii, iii and iv

CED172

39. In which of the following pairs first one is having more resonance energy than the second one -

(A) 
(B) 
(C) 
(D) None of these

CED173

For More Join: @IITJEE_Advanced

Hyperconjugation

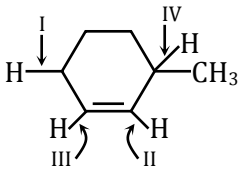
40. Which one of the following molecules has all the effect, namely inductive, mesomeric and hyper conjugative?

(A) CH_3Cl (B) $CH_3-CH=CH_2$

(C) $CH_3-CH=CH-\underset{\underset{O}{\|}}{C}-CH_3$ (D) $CH_2=CH-CH=CH_2$

CED189

41. Which of the following C-H bonds participate in hyper conjugation?



(A) I and II (B) I and IV (C) I and III (D) III and IV

CED190

42. Hyperconjugation is possible in which of the following species?

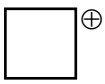
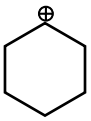
(A) $CH_3-\overset{\ominus}{C}H-CH_3$ (B) $C_6H_5-CH_3$ (C) $H_2C=CH_2$ (D) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH=CH_2$

Rankers Academy JEE

CED242

Bayers strain theory

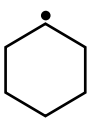
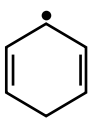
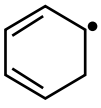
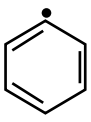
43. Among given carbocations, which is most stable

(A)  (B)  (C)  (D) 

CED196

Applications of Electronic displacement effect; Stability of Intermediates and rearrangement of Carbocation

44. The increasing order of stability of the following free radicals is

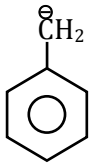
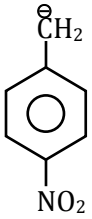
(i)  (ii)  (iii)  (iv) 

(A) I > II > III >IV (B) III > II > I >IV (C) IV > III > II >I (D) I > III > II >IV

CED129

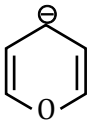
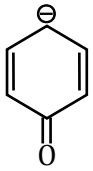
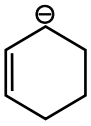
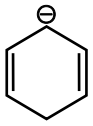
For More Join: @IITJEE_Advanced

45. Most stable carbanion is :

(A) $CH_3^{\ominus}$ (B) $CH_2=CH-\overset{\ominus}{C}H_2$ (C)  (D) 

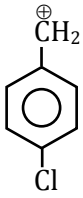
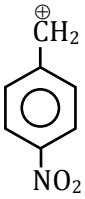
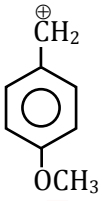
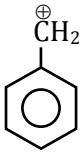
CED130

46. Identify the most stable anion.

(A)  (B)  (C)  (D) 

CED131

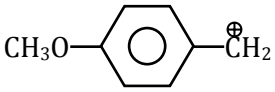
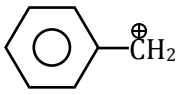
47. Most stable carbocation is :-

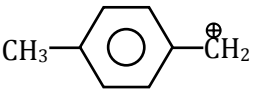
(A)  (B)  (C)  (D) 

CED191

Rankers Academy JEE

48. Consider the following carbocations

(a)  (b) 

(c)  (d) $CH_3-\overset{\oplus}{C}H_2$

The relative stabilities of these carbocations are such that :-

(A) d < b < c < a (B) b < d < c < a (C) d < b < a < c (D) b < d < a < c

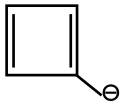
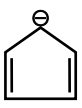
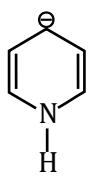
CED192

49. Among the following ions, which is most stable.

(A) $Ph-\overset{\ominus}{C}H_2$ (B) $H_2C=CH-\overset{\ominus}{C}H_2$ (C) $\overset{\ominus}{O}H$ (D) 

CED195

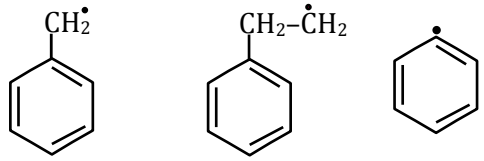
50. Among the following most stable carbanion is

(A)  (B)  (C)  (D) 

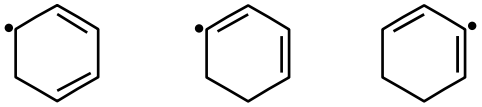
CED199

For More Join: @IITJEE_Advanced

51. Correct order of stability for given intermediates is :

(i) 

(p) (q) (r)

(ii) 

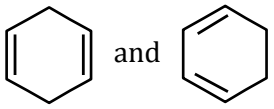
(x) (y) (z)

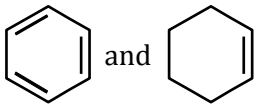
(A) p > r > q ; x > y > z (B) p > q > r ; z > y > x
(C) p > q > r ; x > y > z (D) r > p > q ; y > x > z

CED201

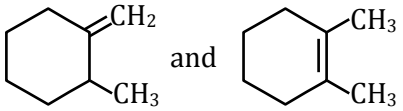
Heat of hydrogenation

52. Among the following pairs, identify the one in which second compound gives higher heat of hydrogenation:

(A)  and

(B)  and

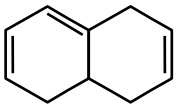
(C) $CH_3–CH=CH–CH_3$ and $CH_3–CH_2–CH=CH_2$

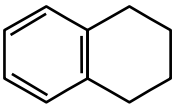
(D) 

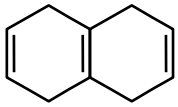
CED203

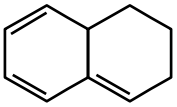
Rankers Academy JEE

53. Compound having highest heat of hydrogenation is:

(A) 

(B) 

(C) 

(D) 

CED204

Heat of combustion

54. Rank the following alkenes in decreasing order of heat of combustion values :

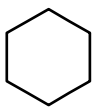
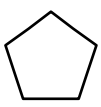


(I) (II) (III) (IV)

(A) II > III > IV > I (B) II > IV > III > I (C) I > III > IV > II (D) I > IV > III > II

CED202

55. Give decreasing order of heat of combustion (HOC):

(a)  (b)  (c)  (d) 

(A) a > b > c > d (B) a > b > d > c (C) c > b > d > a (D) d > b > a > c

CED206

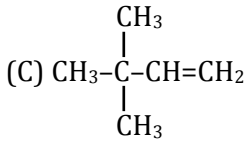
For More Join: @IITJEE_Advanced

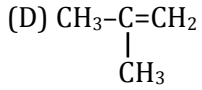
Bond length and Bond order

56. Which of the following molecule has longest C=C bond length?

(A) $CH_2=C=CH_2$

(B) $CH_3-CH=CH_2$

(C) 

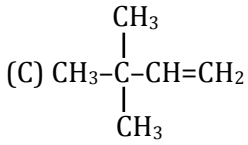
(D) 

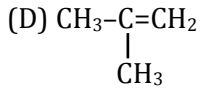
CED170

57. Which of the following molecule has shortest C=C bond length?

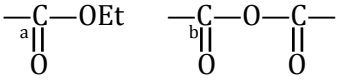
(A) $CH_2=C=CH_2$

(B) $CH_3-CH=CH_2$

(C) 

(D) 

CED171

58. **Statement-I :**  bond length a < b

Because

Statement-II : More is the double bond character less is the bond length.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

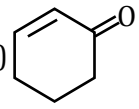
(C) Statement-1 is true, statement-2 is false.

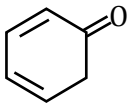
(D) Statement-1 is false, statement-2 is true.

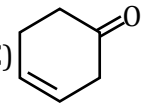
Paakets Academy JEE

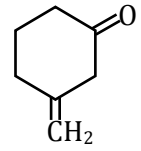
CED178

59. Which of the following has longest C–O bond :

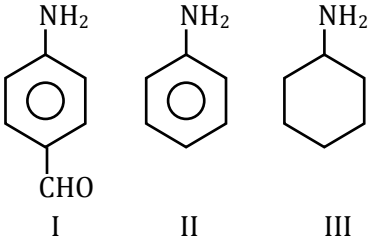
(A) 

(B) 

(C) 

(D) 

CED179

60. 

I II III IV

Among these compounds, the correct order of C–N bond lengths is:

(A) IV > I > II > III

(B) III > I > II > IV

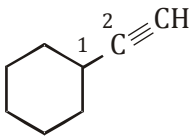
(C) III > II > I > IV

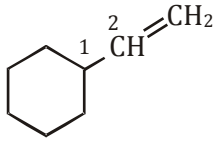
(D) III > I > IV > II

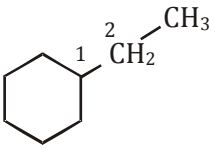
CED180

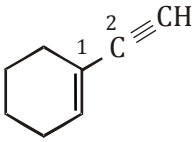
For More Join: @IITJEE_Advanced

61. C_1—C_2 bond is shortest in

(A) 

(B) 

(C) 

(D) 

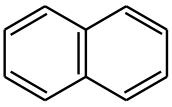
CED181

62. C—C and C=C bond lengths is equal in:

(A) Benzene

(B) 1,3-buta-di-ene

(C) 1,3-cyclohexa-di-ene

(D) 

CED182

63. Among the following molecules, the correct order of C – C bond length is (C_6H_6 is benzene)

(A) $C_2H_6 > C_2H_4 > C_6H_6 > C_2H_2$

(B) $C_2H_6 > C_6H_6 > C_2H_4 > C_2H_2$

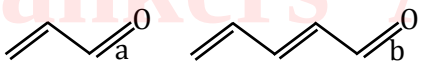
(C) $C_2H_4 > C_2H_6 > C_2H_2 > C_6H_6$

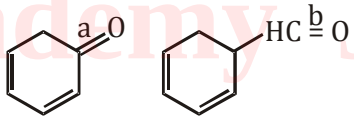
(D) $C_2H_6 > C_2H_4 > C_2H_2 > C_6H_6$

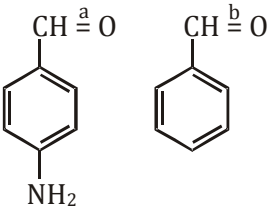
CED184

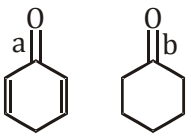
64. In which case, C–O bond length is shorter for I[st] compound:

Rankers Academy JEE

(A) 

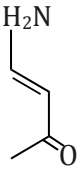
(B) 

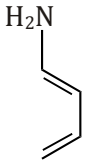
(C) 

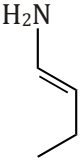
(D) 

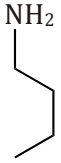
CED186

65. C–N bond length is maximum in:

(A) 

(B) 

(C) 

(D) 

CED187

For More Join: @IITJEE_Advanced

Acidic strength

66. Give correct order of acidic strength of following compounds :

(a) $NO_2-CH_2-\overset{\overset{O}{\|}}{C}-O-H$

(b) $F-CH_2-\overset{\overset{O}{\|}}{C}-O-H$

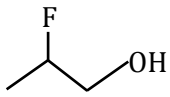
(c) $Ph-CH_2-\overset{\overset{O}{\|}}{C}-O-H$

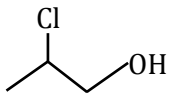
(d) $CH_3-CH_2-\overset{\overset{O}{\|}}{C}-O-H$

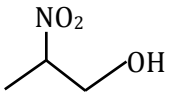
(A) a > b > c > d (B) a > c > b > d (C) b > a > c > d (D) d > c > a > b

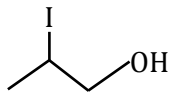
CED110

67. Which of the following is strongest acid.

(A) 

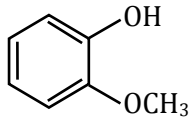
(B) 

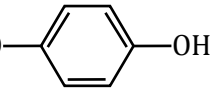
(C) 

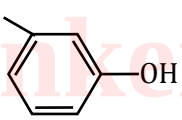
(D) 

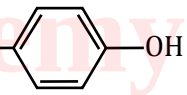
CED113

68. In the following compounds, the order of acidic strength is

(I) 

(II) H_3CO  OH

(III) 

(IV) O_2N  OH

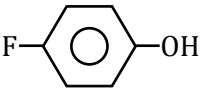
Rankers Academy JEE

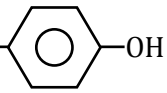
(A) IV > I > III > II (B) IV > II > III > I (C) IV > III > II > I (D) IV > III > I > II

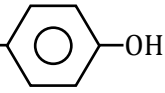
CED209

69. Arrange the given phenol & its derivative in their decreasing order of acidity :

(I) C_6H_5-OH

(II) F  OH

(III) Cl  OH

(IV) O_2N  OH

Select the correct answer from the given code:

(A) IV > III > I > II (B) IV > II > III > I (C) IV > III > II > I (D) IV > I > III > II

CED210

70. Which one of the following has the most acidic hydrogen ?

(A) 

(B) 

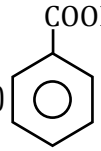
(C) 

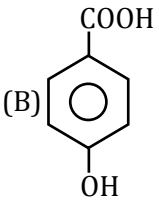
(D) $CH_2=CH-CH_3$

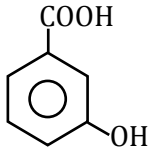
CED211

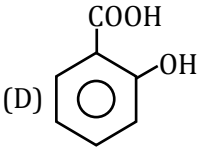
For More Join: @IITJEE_Advanced

71. Which of the following is weakest acid?

(A) 

(B) 

(C) 

(D) 

CED212

72. The strongest acid is :

(A) HF (B) CH_3CO_2H (C) $HF + SbF_5$ (D) H_2S

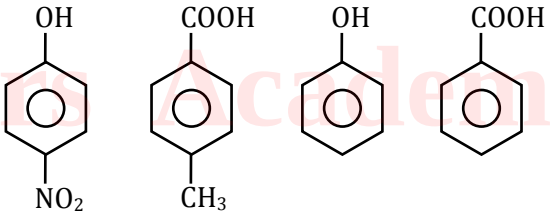
CED214

73. The weakest acid (does not show acidic character) is :

(A) $HC \equiv CH$ (B) $CH_2 = CH_2$ (C) Me_3CH (D) Ph_3CH

CED215

74. Acidic strength order of given compound

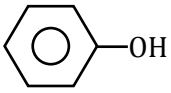


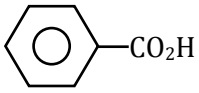
(I) (II) (III) (IV)

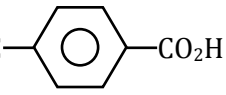
(A) I > IV > III > II (B) I > IV > II > III (C) IV > II > I > III (D) IV > I > II > III

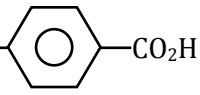
CED217

75. Most acidic compound is :-

(A) 

(B) 

(C) 

(D) 

CED218

76. The correct order of acidic strength of the following compounds is :-

A. Phenol B. p-Cresol

C. m-Nitrophenol D. p- Nitrophenol

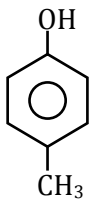
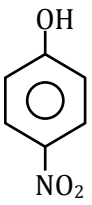
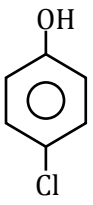
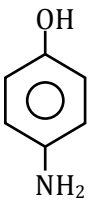
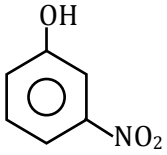
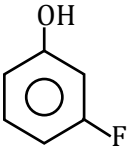
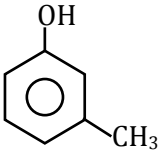
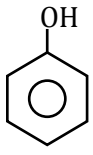
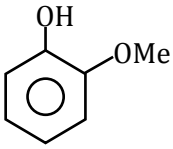
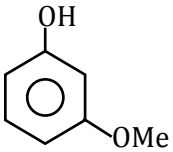
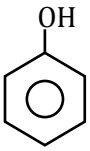
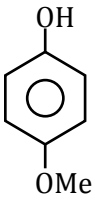
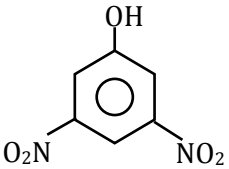
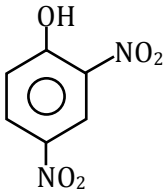
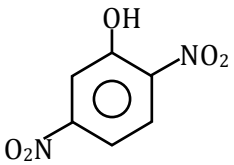
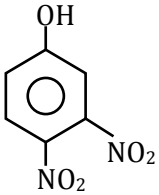
(A) C > B > A > D (B) D > C > A > B

(C) B > D > A > C (D) A > B > D > C

CED219

For More Join: @IITJEE_Advanced

77. Select the strongest acid in each of the following sets:

(i)				
	(a)	(b)	(c)	(d)
(ii)				
	(a)	(b)	(c)	(d)
(iii)				
	(a)	(b)	(c)	(d)
(iv)				
	(a)	(b)	(c)	(d)

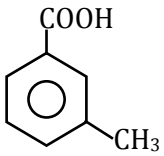
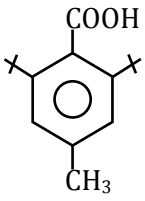
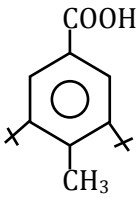
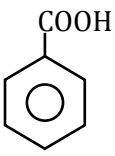
(A) (i) → b ; (ii) → a ; (iii) → b ; (iv) → b
(B) (i) → a ; (ii) → b ; (iii) → c ; (iv) → b
(C) (i) → c ; (ii) → c ; (iii) → a ; (iv) → a
(D) (i) → d ; (ii) → d ; (iii) → c ; (iv) → d

CED224

Rankers Academy JEE

SIR effect; Ortho effect and Mixed questions

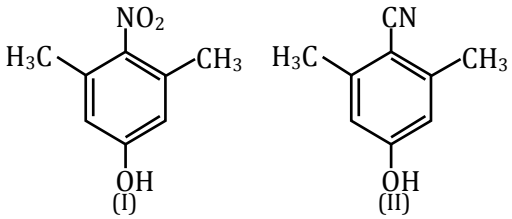
78. Which of the following is most acidic:

(A) 
(B) 
(C) 
(D) 

CED223

For More Join: @IITJEE_Advanced

79. Statement-1 : For the given two compounds-I is more acidic than compounds-II.

and 

Statement-2 : Due to presence of $-CH_3$ group at ortho positions to $-NO_2$; the plane of $-NO_2$ deviates, w.r.t plane of ring.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
(C) Statement-1 is True, Statement-2 is False.
(D) Statement-1 is False, Statement-2 is True.

CED227

Electrophile, Nucleophile, solvent, leaving group, Nucleophilicity vs Basicity

80. An 'Electrophile' from the following is-

(A) SO_2 (B) NH_3 (C) CO_2 (D) A & C

CED281

81. An electrophile from the following is-

(A) $FeCl_3$ (B) $I^{\ominus}$ (C) $\overset{\ominus}{O}H$ (D) None of these

CED282

Rankers Academy JEE

82. 'A' Neucleophile from the following is-

(A) $AlCl_3$ (B) BCl_3 (C) NH_3 (D) $Cl^{\oplus}$

CED283

83. The least stable carbanion from the following is-

(A) $CH_3^{\ominus}$ (B) $CH_2=CH-CH_2^{\ominus}$ (C) $(CH_3)_3C^{\ominus}$ (D) $(CH_3-CH_2)_3C^{\ominus}$

CED284

84. The electrophile from the following is-

(A) $NH_4^{\oplus}$ (B) $H_3O^{\oplus}$ (C) $H^{\oplus}$ (D) None of these

CED285

85. Which of the following compound acts as both Electrophile and Nucleophile-

(A) C_2H_5OH (B) CO_2 (C) CH_3-CH_2-CN (D) None of these

CED286

86. Arrange the following as increasing order of their ability as leaving group:-

$^{\ominus}OTs$ (Q), $CF_3^{\ominus}$ (R), & $F^{\ominus}$(S)

(A) S < Q < R (B) Q < R < S (C) R < Q < S (D) R < S < Q

CED287

87. Which of the following is a non-polar solvent?

(A) H_2O (B)  (C) CCl_4 (D) Ethanol

CED288

For More Join: @IITJEE_Advanced

88. The carbon free radicals are :

(A) diamagnetic (B) Paramagnetic

(C) Ferromagnetic (D) Anti Ferromagnetic

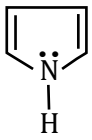
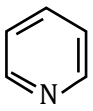
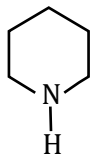
CED289

89. The maximum nucleophilicity is in which of the followings:

(A) $CH_3-\overset{\ominus}{C}H_2$ (B) $CH_2=CH-\overset{\ominus}{C}H_2$ (C) $CH\equiv C-\overset{\ominus}{C}H_2$ (D) None of these

CED290

90. Which of the following is better Nucleophile :-

(A)  (B)  (C)  (D) None of these

CED291

91. Among the following nucleophiles, which of the following order is correct for their nucleophilicity in water?

(A) $\overset{\ominus}{C}H_3 > \overset{\ominus}{N}H_2 > \overset{\ominus}{O}H > \overset{\ominus}{F}$ (B) $\overset{\ominus}{F} > \overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{C}H_3$

(C) $\overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{F} > \overset{\ominus}{C}H_3$ (D) $\overset{\ominus}{C}H_3 > \overset{\ominus}{O}H > \overset{\ominus}{N}H_2 > \overset{\ominus}{F}$

CED292

92. The decreasing order of nucleophilicities of the following is:

(I) H_2O (II) EtOH (III) $MeCOO^{\ominus}$ (IV) $^{\ominus}OH$

(V) $EtO^{\ominus}$

(A) (V) > (IV) > (III) > (II) > (I) (B) (I) > (II) > (III) > (IV) > (V)

(C) (IV) > (V) > (III) > (II) > (I) (D) (I) > (II) > (III) > (V) > (IV)

CED293

Rankers' Academy JEE

Basic strength + SIP Effect

93. Give decreasing order of basic strength of following:

(a) $CH_3^{\ominus}$ (b) $NH_2^{\ominus}$ (c) $OH^{\ominus}$ (d) $F^{\ominus}$

(A) a > b > c > d (B) a > c > b > d (C) b > a > c > d (D) d > c > a > b

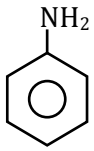
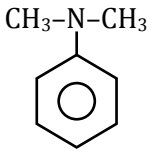
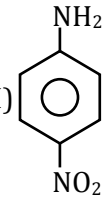
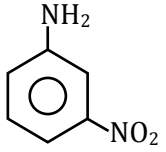
CED109

94. Which among the following is more basic?

(A) $HC\equiv C^{\ominus}$ (B) $C_6H_5^{\ominus}$ (C) $(CH_3)_3C-CH_2^{\ominus}$ (D) $(CH_3)_2C=CH^{\ominus}$

CED111

95. Arrange the following in the decreasing order of basicity:

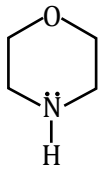
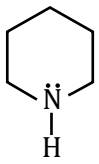
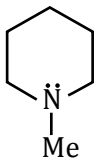
(I)  (II)  (III)  (IV) 

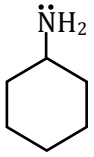
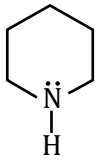
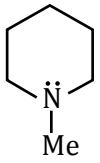
(A) I > II > III > IV (B) IV > III > II > I (C) II > I > IV > III (D) I > IV > III > II

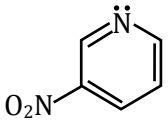
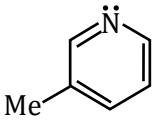
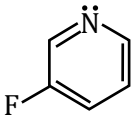
CED229

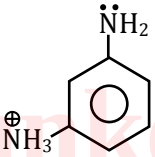
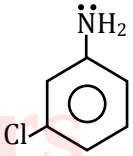
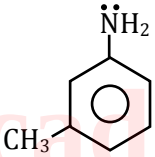
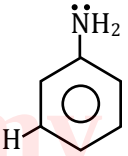
For More Join: @IITJEE_Advanced

96. Correct set of basic strength order in given options is:

(A)  (a)  (b)  (c) (c > b > a)

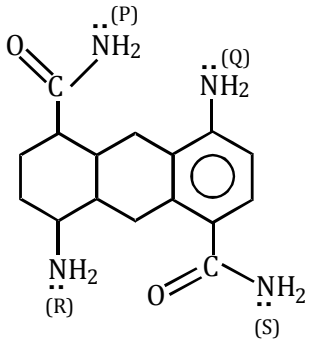
(B)  (a)  (b)  (c) (b > a > c)

(C)  (a)  (b)  (c) (a > b > c)

(D)  (a)  (b)  (c)  (d) (d > b > a > c)

Rankers Academy JEE

CED230

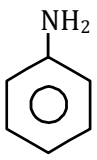
97. 

The mostbasic N-atom is :

(A) P (B) Q (C) R (D) S

CED231

98. Which compound is most basic -

(A) $CH_3—CH_2—NH_2$ (B) $CH_3—CH=NH$ (C) $CH_3—C\equiv N$ (D) 

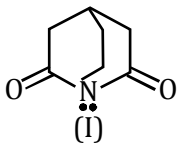
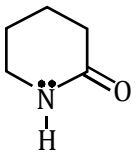
CED232

For More Join: @IITJEE_Advanced

99. Choose the strongest base among the following

(A) $C_6H_5CONHCH_3$ (B) $C_6H_5NH_2$ (C) $C_6H_5–CO–NH_2$ (D) $C_6H_5–NH–CH_3$

CED238

100. Statement 1 :  is more basic than  and

(I) (II)

Statement 2 : Lone pair electrons on nitrogen in compound (I) does not participate in resonance.

(A) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.

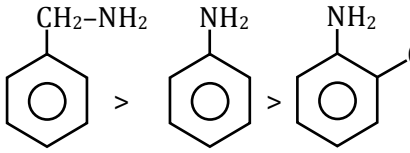
(B) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.

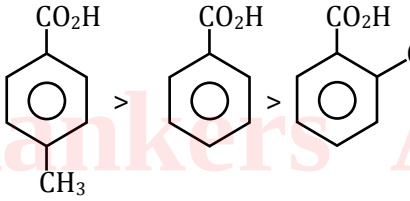
(C) Statement-1 is True, Statement-2 is False.

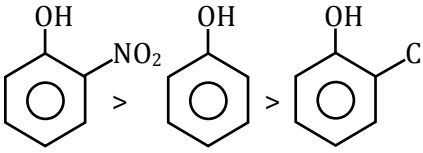
(D) Statement-1 is False, Statement-2 is True.

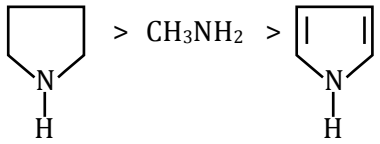
CED241

101. What is the incorrect order.

(A)  (Basic strength)

(B)  (Acidic strength)

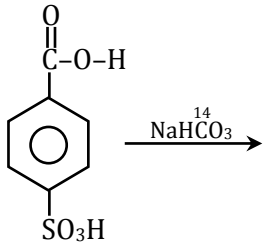
(C)  (Acidic strength)

(D)  (Basic strength)

CED233

Acid Base Equilibrium; $NaHCO_3$ test and Reaction of Na, aq. NaOH, 5% HCL

102. In given reaction Gas liberated is/are

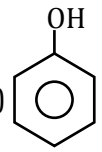


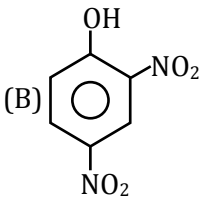
(A) CO_2 & SO_2 (B) SO_2 & $^{14}CO_2$ (C) $^{14}CO_2$ only (D) SO_2 only

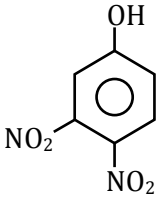
CED236

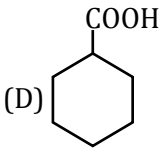
Rankers Academy JEE
For More Join: @IITJEE_Advanced

103. Which of the following compound does not react with $NaHCO_3$

(A) 

(B) 

(C) 

(D) 

CED237

Purification of Organic Compounds, Sublimation, Crystallization, Distillation, Steam Distillation, Differential Extraction

104. For the Separation of two immiscible liquids which method (or apparatus) is used?

(A) Chromatography
(B) Fractionating column
(C) Fractional distillation
(D) Separating funnel / Decantation

CED294

105. Aniline is usually purified by

(A) Chromatographic technique
(B) Steam distillation
(C) By addition of oxalic acid
(D) Fractional crystallization

CED295

106. The most suitable method for separation of 1:1 mixture of ortho- and para- nitrophenols is

(A) Sublimation
(B) Chromatography
(C) Crystallization
(D) Steam distillation

CED296

Rankers Academy JEE

107. Distillation under reduced pressure is employed for:

(A) Benzene
(B) Petrol
(C) Glycerol
(D) Organic compound used in medicine

CED297

Chromatography

108. The latest technique used for purification for organic compounds is

(A) Chromatography
(B) Sublimation
(C) Fractional distillation
(D) Crystallisation

CED298

109. In paper chromatography

(A) Mobile phase is liquid and stationary phase is solid
(B) Mobile phase is liquid and stationary phase is liquid
(C) Mobile phase is solid and stationary phase is solid
(D) Mobile phase is solid and stationary phase is liquid

CED299

Element Detection; Detection of Carbon and Hydrogen; Detection of Nitrogen, Sulphur, Halogens and Phosphorus; Lassaigne's Test; Beilstein Test

110. In sodium fusion test of organic compound, the nitrogen of organic compound is converted into

(A) Sodamide
(B) Sodium cyanide
(C) Sodium nitrite
(D) Sodium nitrate

CED300

For More Join: @IITJEE_Advanced

111. The Lassaigne's extract is boiled with dil. HNO_3 before testing for halogens because

(A) Silver halides are soluble in HNO_3 (B) Na_2S and NaCN are decomposed by HNO_3

(C) Ag_2S is soluble in HNO_3 (D) AgCN is soluble in HNO_3

CED301

112. In a Lessaigne's test for nitrogen, the blue colour is due to the formation of

(A) Potassium ferricyanide (B) Sodium cyanide

(C) Sodium ferrocyanide (D) Ferri-ferro cyanide

CED302

113. Which of the following compounds does not show Lassaigne test for nitrogen?

(A) Urea (B) Hydrazine

(C) Phenyl hydrazine (D) Azo benzene

CED303

114. The Beilstein test for organic compounds is used to detect

(A) Nitrogen (B) Sulphur (C) Carbon (D) Halogen

CED304

115. Which of the following is the best scientific method to test presence of water in liquid?

(A) Smell (B) Taste

(C) Use of litmus paper (D) Use of anhydrous copper sulphate

CED305

Quantitative Analysis of Organic Compounds (Kjeldahl's method, Dumas method, Carius method)

Rankers Academy JEE

116. In Kjeldahl's method, nitrogen present is estimated as

(A) N_2 (B) NO (C) NH_3 (D) NO_2

CED306

117. Liebig's method is used for the estimation of

(A) Nitrogen (B) Sulphur

(C) Carbon and hydrogen (D) Halogens

CED307

118. 0.765 gm of an acid gives 0.535 gm of CO_2 and 0.138 gm of H_2O. Then the ratio of percentage of carbon and hydrogen is

(A) 19 : 2 (B) 18 : 11 (C) 20 : 17 (D) 1 : 7

CED308

119. Nitrogen is an organic compound can be estimated by

(A) Kjeldahl's method only (B) Duma's method only

(C) Both the methods (D) None of these methods

CED309

120. Dumas method involves the determination of nitrogen content in the organic compound in form of

(A) NH_3 (B) N_2 (C) NaCN (D) $(NH_4)_2SO_4$

CED310

For More Join: @IITJEE_Advanced

EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

Mesomeric effect (Groups showing +M and -M effects)

1. Which of the following group(s) is/are ortho and para directing?

(A) –CN (B) $-SO_3H$ (C) $-NH_2$ (D) 

CED140

2. Which of the following have + M effect ($e^{\ominus}$–donating mesmeric effect)?

(A) $-NO_2$ (B) –COOH (C) $-NH_2$ (D) –SR

CED141

3. Which of the following have –M effect?

(A)  (B) $-SO_3H$ (C) –OR (D) –Br

CED142

4. Which of the following groups show both +M and –M effect depending upon situation.

(A) $-NO_2$ (B) –NO (C) $-CH=CH_2$ (D) –CHO

CED143

5. Which of the following group can neither show +M are –M effect?

(A) $-\overset{\oplus}{P}H_3$ (B) $-\overset{\oplus}{N}H_3$ (C) –OH (D) $-CH_3$

CED144

Rankers Academy JEE

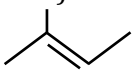
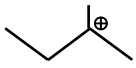
6. Group which can show both –I and +M effects:

(A) OH (B) –COCl (C) –CHO (D) –Cl

CED145

Hyperconjugation

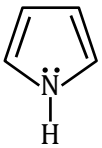
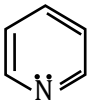
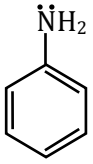
7. Which of the following compound is correctly matched with number of hyper conjugating structures (involving C–H bond) :

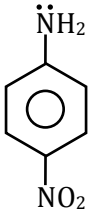
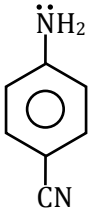
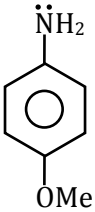
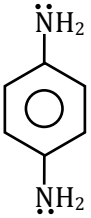
(A)  (3) (B)  (9) (C)  (8) (D) $CH_3-C\equiv C-CH_3$ (5)

CED207

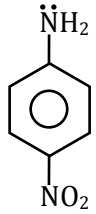
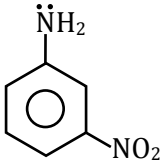
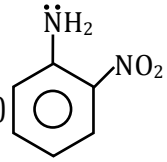
Basic strength + SIP Effect

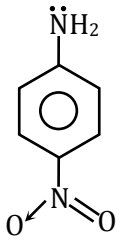
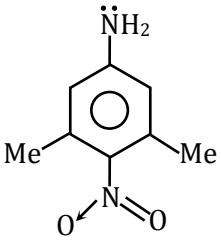
8. Correct set of basic strength order in given option is/are:

(A) (a)  (b)  (c)  (b > c > a)

(B) (a)  (b)  (c)  (d)  (d > c > b > a)

For More Join: @IITJEE_Advanced

(C) (a)  (b)  (c)  (b > a > c)

(D) (a)  (b)  (a > b)

CED244

Acid Base Equilibrium; $NaHCO_3$ test and Reaction of Na, aq. NaOH, 5% HCL

9. Identify the non-feasible reaction

(A) $CH_3-C\equiv CH+NH_2^{\ominus} \rightleftharpoons CH_3-C\equiv \ddot{C}^{\ominus}+NH_3$

(B) $CH_3CH_2-OH+NaOH \rightleftharpoons CH_3CH_2ONa+H_2O$

(C) $CH_3-OH+NaOH \rightleftharpoons CH_3ONa+H_2O$

(D) $HC\equiv CH+NaOH \rightleftharpoons HC\equiv CNa+H_2O$

CED243

Rankers Academy JEE

Quantitative Analysis of Organic Compounds (Kjeldahl's method, Dumas method, Carius method)

Passage for Question 10 to 14

The 0.2 g of anhydrous organic acid gave on combustion 0.04 g of water and 0.195 g of CO_2, the acid is a dibasic acid and 0.5 g of its silver salt leaves on ignition 0.355 g of silver.

10. The percentage of carbon in the compound is

(A) 50 (B) 52 (C) 26.6 (D) 90

CED311

11. The percentage of hydrogen in the compound is

(A) 5.6 (B) 2.22 (C) 4.44 (D) 10

CED312

12. The empirical weight of the compound is

(A) 90 (B) 100 (C) 10 (D) 45

CED313

13. The molecular weight of the compound is

(A) 90 (B) 100 (C) 10 (D) 45

CED314

14. The molecular formula of the compound is

(A) CHO (B) CHO_2 (C) $C_2H_2O_4$ (D) $C_3H_6O_2$

CED315

For More Join: @IITJEE_Advanced

EXERCISE (S)

Part-1 : JEE Main (Numerical Value type)

- The answer to each question is a **Numerical Value**.
- For each question, enter the correct integer value (In case of non-integer value, the answer should be **rounded off** to the nearest Integer).

Inductive effect (Groups showing +I and –I effects)

1. How many of the following groups have + I effect :

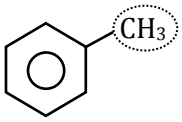
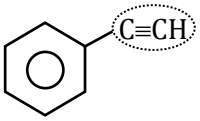
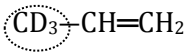
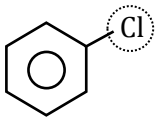
(a) –OH (b) $-O^{\ominus}$ (c) $-NH_2$ (d) $-\overset{\ominus}{N}H$

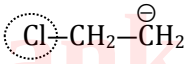
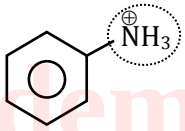
(e) –COOH (f) $-COO^{\ominus}$ (g) –Me (h) –OMe

(i) –F (j) $-N\overset{\oplus}{F}_3$

CED108

2. Calculate the marked group which shows–I effect.

(i)  (ii)  (iii)  (iv) 

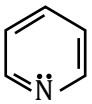
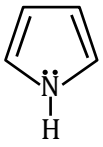
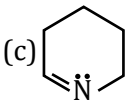
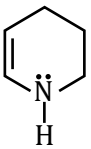
(v)  (vi) $H_2C{=}CH{-}C{\equiv}CH$ (vii) 

Rankers Academy JEE

CED269

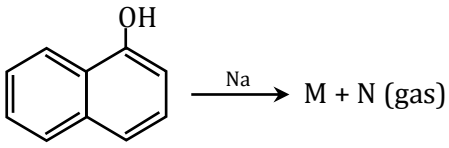
Resonance, Conditions for resonance, Stability of Resonating structures

3. Number of compounds In which lone-pair indicated is involved in resonance:

(a)  (b)  (c)  (d) 

(e) $CH_2=CH-\overset{\ominus}{C}H_2$ (f) $CH_2=CH-CH=\ddot{N}H$

CED128

4. 

CED270

Mesomeric effect (Groups showing +M and -M effects)

5. How many of these groups shows –R effect when attached to benzene :

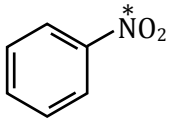
(a) $-NO_2$ (b) $-CH_3$ (c) $-OCH_3$ (d) $-COCH_3$

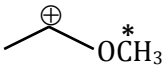
(e) $-NH_2$ (f) $-CONH_2$

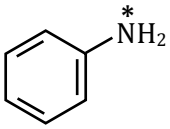
CED146

For More Join: @IITJEE_Advanced

6. How many compounds having * marked group show both –I & + R / + M effects.

(a) $CH_3–CH=CH–\overset{*}{C}HO$ (b) 

(c) $H_2C=CH–\overset{*}{F}$ (d) 

(e) $H_3C–CH=CH–\overset{*}{C}H_3$ (f) 

CED147

7. How many species is/are –M/–R groups

(i) –OH (ii) $-NO_2$ (iii) –CN (iv) $-CH_3$

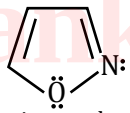
(v) $-O^{\ominus}$ (vi) $-\overset{\oplus}{N}H_3$ (vii) –COOH (viii) –CH=NH

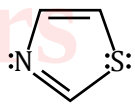
CED148

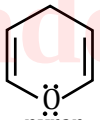
Aromaticity

8. Among the given compounds, total number of non-aromatic species are :

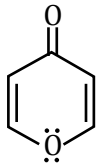
Rankers Academy JEE

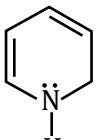
(a)  isoxazole

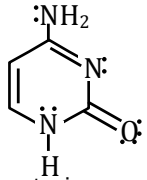
(b)  1,3-thiazole

(c)  pyran

(d)  Pyrylium ion

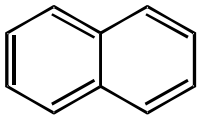
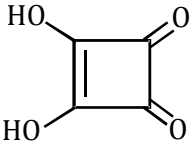
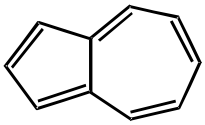
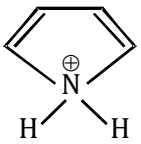
(e)  γ–pyrone

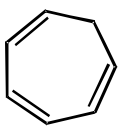
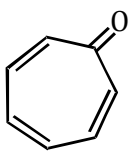
(f)  1,2–dihydropyridine

(g)  cytosine

CED167

9. Total number of aromatic compounds

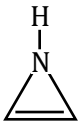
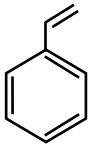
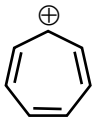
(a)  (b)  (c)  (d) 

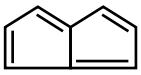
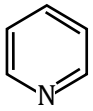
(e)  (f) 

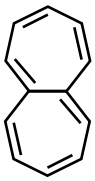
CED168

For More Join: @IITJEE_Advanced

10. Of the given system how many molecule are anti-aromatic ?

(a)  (b)  (c)  (d) 

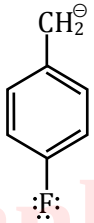
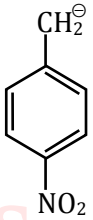
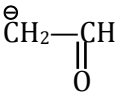
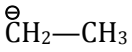
(e)  (f)  (g)  (h) 

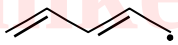
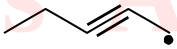
(i)  (j) 

CED280

Hyperconjugation

11. Number of pairs having first intermediate more stable is:

(i) (a)  (b)  (ii) (a)  (b) 

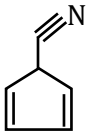
(iii) (a)  (b) 

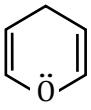
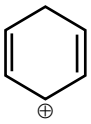
Rankers Academy JEE

CED208

Acidic strength

12. How many compound are stable after deprotonation?

(a)  (b)  (c)  (d) 

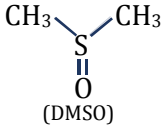
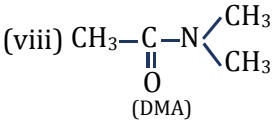
(e)  (f) 

CED271

Electrophile, Nucleophile, solvent, leaving group, Nucleophilicity vs Basicity

13. How many of the following are polar protic solvents :

(i) CCl_4 (ii) Benzene (iii) H_2O (iv) CH_3CH_2OH

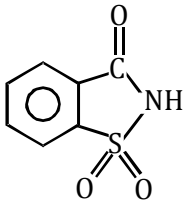
(v) CH_3CH_2OH (vi) NH_3 (vii)  (DMSO) (viii)  (DMA)

CED316

For More Join: @IITJEE_Advanced

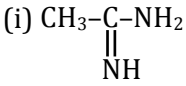
Basic strength + SIP Effect

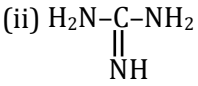
14. The structure of saccharin is given as follows :

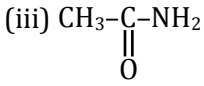


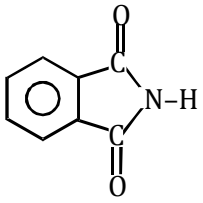
[Provided N is the donor atom in all cases]

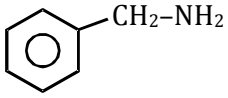
How many following compounds are more basic than saccharin?

(i) 

(ii) 

(iii) 

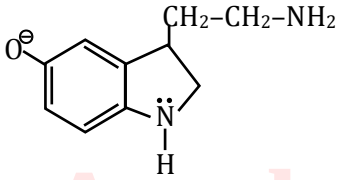
(iv) 

(v) 

(vi) 

CED246

15. The conjugate base of serotonin (used as tranquilisers) is given as follows:

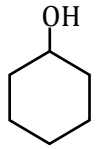


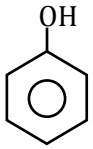
How many basic groups present in following compound?

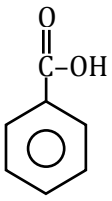
CED248

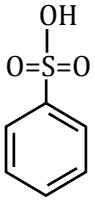
Acid Base Equilibrium; $NaHCO_3$ test and Reaction of Na, aq. NaOH, 5% HCL

16. How many following compound will react with $NaHCO_3$ or soluble in $NaHCO_3$?

(a) 

(b) 

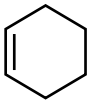
(c) 

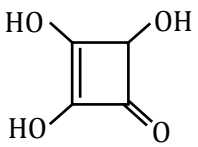
(d) 

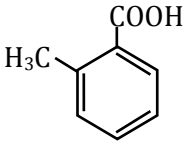
(e) $CH_3-C\equiv CH$

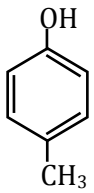
CED272

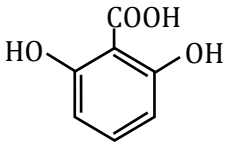
17. How many compounds are soluble aqueous NaOH?

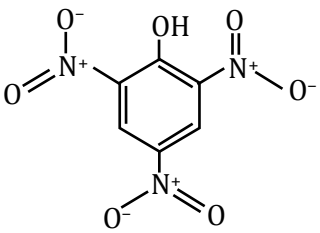
(a) 

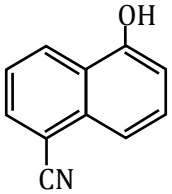
(b) 

(c) 

(d) 

(e) 

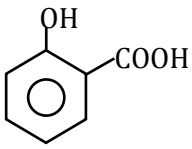
(f) 

(g) 

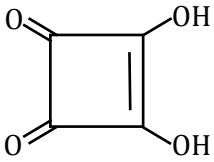
CED273

Rankers Academy JEE
For More Join: @IITJEE_Advanced

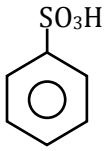
18. Number of compounds which can give effervescences with $NaHCO_3$

(i) 

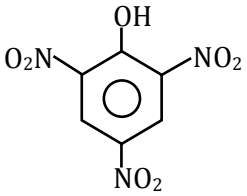
(Salicylic acid)

(ii) 

(Squaric acid)

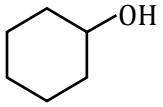
(iii) 

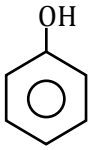
(Benzene sulphonic acid)

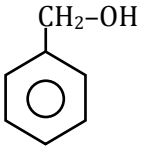
(iv) 

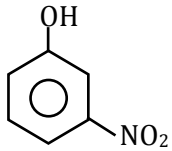
(Picric acid)

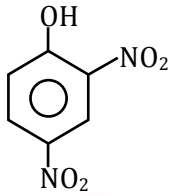
(v) Ph–CH=CH–COOH
(Cinnamic acid)

(vi) 

(vii) 

(viii) 

(ix) 

(x) 

CED247

19. Total number of compounds soluble in aqueous HCl :

(a) $R-NH_2$ (b) R_2NH (c) R_3N (d) $R-\overset{\overset{O}{\|}}{C}-NH_2$

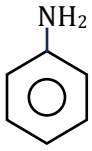
(e) PhOH (f) $PhNH_2$ (g) $Ph-COOH$ (h) $PhCH_3$

(i) $Ph-CH_2-NH_2$

CED274

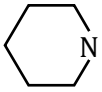
Element Detection; Detection of Carbon and Hydrogen; Detection of Nitrogen, Sulphur, Halogens and Phosphorus; Lassaigne's Test; Beilstein Test

20. Amongst the following, the total number of compounds which does not give Lassaigne's test for nitrogen.

I. 

II. H_2N-CH_2-COOH III. NH_2-NH_2 IV. $Ph-N=N-Ph$ V. NH_2OH

VI. $NH_2-\overset{\overset{O}{\|}}{C}-NH_2$ VII. $Ph-NH-\overset{\overset{O}{\|}}{C}-Ph$ VIII. $PhNO_2$ IX. CH_3CHO X. CH_3CN

XI. 

CED317

For More Join: @IITJEE_Advanced

21. Amongst the following compounds the total number of compounds which give Beilstein test.

I. Urea **II.** Thiourea **III.** Pyridine **IV.** CH_3F
V. CH_3Cl **VI.** CH_3Br **VII.** CH_3I **VIII.** Benzamide
IX. Benzaldehyde **X.** Aniline

CED318

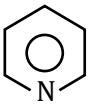
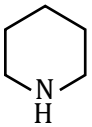
22. Amongst the following, the total number of compounds which are expected to give blood red colouration when $FeCl_3$ is added to their respective Lassaigne's extract is:

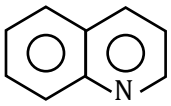
(1) thiourea (2) benzenesulphonic acid (3) sulphanilic acid
(4) diphenyl ulphide (5) urea (6) benzanilide
(7) semicarbazide hydrochloride (8) *p*-toluidine (9) *m*-dinitrobenzene

CED319

Quantitative Analysis of Organic Compounds (Kjeldahl's method, Dumas method, Carius method)

23. Amongst the following Kjeldahl's method can't be used for.

I. $PhNO_2$ II. Ph–N=N–Ph III. PhNHCOPh IV.  V. 

VI.  VII. $PhNH_2$ VIII. Acetone IX. Ethane nitrile X. Urea

CED320

24. 0.50 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.5 M H_2SO_4. The residual acid required 60 mL of M/2 NaOH solution. Find the percentage of nitrogen in the compound.

CED321

25. 0.4 g of an organic compound was treated according to Kjeldahl's method. The ammonia evolved was absorbed in 50 mL of 0.5 M H_3PO_3. The residual acid required 30 mL of 0.5 M $Ca(OH)_2$. Find the percentage of N_2 in the compound.

CED322

26. 0.002 g of an organic compound was treated according to Kjeldahl's method. 0.2×10^{-4} mol of H_2SO_4 was required to neutralize NH_3. Calculate the percentage of N

CED323

Part-2 : JEE Advanced (Decimal type)

- The answer to each question is a **NUMERICAL VALUE**.
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.

Quantitative Analysis of Organic Compounds (Kjeldahl's method, Dumas method, Carius method)

27. 0.12 g of an organic compound containing phosphorus gave 0.22 g of $Mg_2P_2O_7$ by the usual analysis. Calculate the percentage of phosphorus in the compound.

CED324

Rankers Academy JEE
For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. A species having carbon with sextet of electrons and can act as electrophile is called

[JEE (Main) 2024]

(1) carbon free radical (2) carbanion

(3) carbocation (4) pentavalent carbon

CED325

2. Ionic reactions with organic compounds proceed through:

(A) Homolytic bond cleavage (B) Heterolytic bond cleavage

(C) Free radical formation (D) Primary free radical

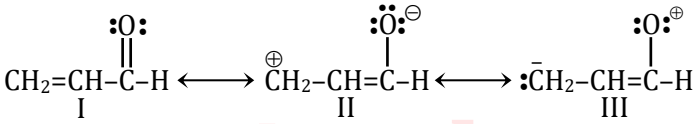
(E) Secondary free radical

Choose the correct answer from the options given below: **[JEE (Main) 2024]**

(1) (A) only (2) (C) only (3) (B) only (4) (D) and (E) only

CED326

3. The order of relative stability of the contributing structure is:

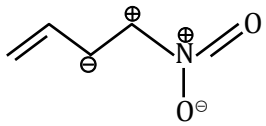


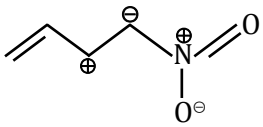
Choose the correct answer from the options given below: **[JEE (Main) 2024]**

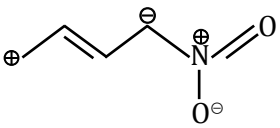
(1) I > II > III (2) II > I > III (3) I = II = III (4) III > II > I

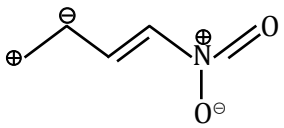
CED327

4. Which one among the following resonating structures is **not** correct ? **[JEE (Main) 2021]**

(1) 

(2) 

(3) 

(4) 

CED042

5. Among the following, total number of meta directing functional groups is (Integer based)

– OCH_3, $-NO_2$, –CN, $-CH_3$ –NHCOCH$_3$,

– COR, –OH, – COOH, –Cl **[JEE (Main) 2024]**

CED328

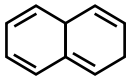
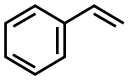
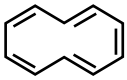
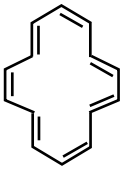
6. The functional group that shows negative resonance effect is: **[JEE (Main) 2024]**

(1) $-NH_2$ (2) –OH (3) –COOH (4) –OR

CED329

For More Join: @IITJEE_Advanced

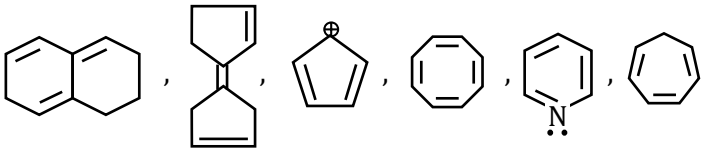
7. Which of the following are aromatic? **[JEE (Main) 2024]**

A.  B.  C.  D. 

(1) B and D only (2) A and C only (3) A and B only (4) C and D only

CED330

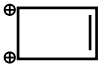
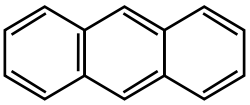
8. Total number of aromatic compounds among the following compounds is _____.



[JEE (Main) 2024]

CED331

9. Which one of the following compounds is non-aromatic? **[JEE (Main) 2021]**

(1)  (2)  (3)  (4) 

CED031

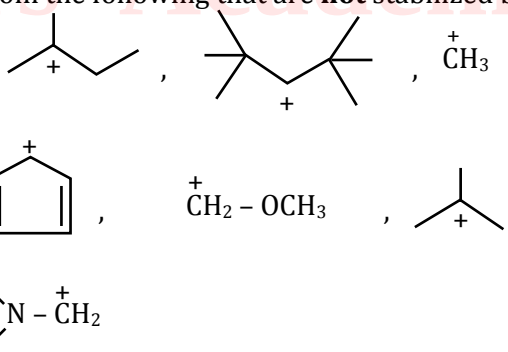
10. The difference in energy between the actual structure and the lowest energy resonance structure for the given compound is **[JEE (Main) 2024]**

(1) electromeric energy (2) resonance energy
(3) ionization energy (4) hyperconjugation energy

CED332

Rankers Academy JEE

11. Number of carbocation from the following that are **not** stabilized by hyperconjugation is..... .



[JEE (Main) 2024]

CED333

12. Given below are two statements: **[JEE (Main) 2021]**

Statement I : Hyperconjugation is a permanent effect.

Statement II : Hyperconjugation in ethyl cation $\left(CH_3-\overset{\oplus}{C}H_2\right)$ involves the overlapping of $C_{sp^2}-H_{1s}$ bond with empty 2p orbital of other carbon.

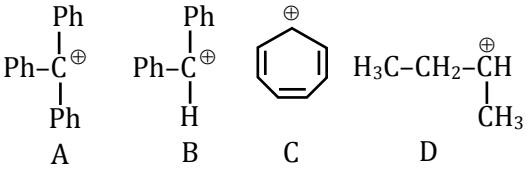
Choose the **correct** option :

(1) Both **statement I** and **statement II** are false
(2) **Statement I** is incorrect but **statement II** is true
(3) **Statement I** is correct but **statement II** is false
(4) Both **Statement I** and **statement II** are true.

CED046

For More Join: @IITJEE_Advanced

13. The correct order of stability of following carbocations is : **[JEE (Main) 2025]**



A B C D

(1) A > B > C > D

(2) B > C > A > D

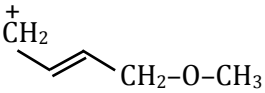
(3) C > B > A > D

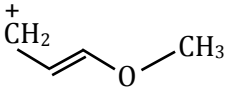
(4) C > A > B > D

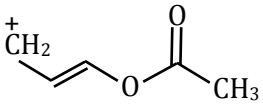
CED373

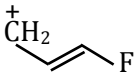
14. Which one of the carbocations from the following is most stable?

[JEE (Main) 2025]

(1) 

(2) 

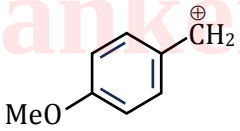
(3) 

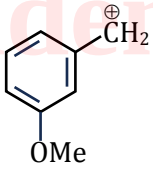
(4) 

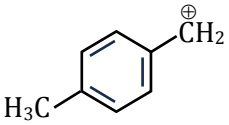
CED374

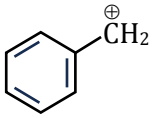
15. The most stable carbocation from the following is: **[JEE (Main) 2025]**

Rankers Academy JEE

(1) 

(2) 

(3) 

(4) 

CED375

16. The correct stability order of carbocations is **[JEE (Main) 2024]**

(1) $(CH_3)_3C^+ > CH_3-\overset{+}{C}H_2 > (CH_3)_2\overset{+}{C}H > \overset{+}{C}H_3$

(2) $\overset{+}{C}H_3 > (CH_3)_2\overset{+}{C}H > CH_3-\overset{+}{C}H_2 > (CH_3)_3\overset{+}{C}$

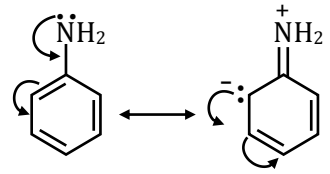
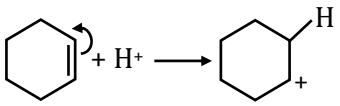
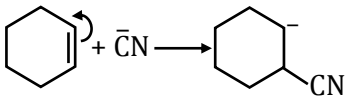
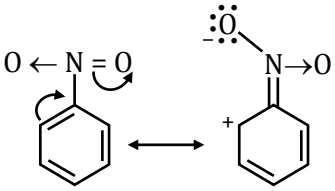
(3) $(CH_3)_3\overset{+}{C} > (CH_3)_2\overset{+}{C}H > CH_3-\overset{+}{C}H_2 > \overset{+}{C}H_3$

(4) $\overset{+}{C}H_3 > CH_3-\overset{+}{C}H_2 > CH_3-\underset{CH_3}{\overset{+}{C}H} > (CH_3)C^+$

CED334

For More Join: @IITJEE_Advanced

17. Match List -I with List II:

List - I Mechanism steps		List - II Effect	
(A)		(I)	– E effect
(B)		(II)	– R effect
(C)		(III)	+ E effect
(D)		(IV)	+ R effect

Choose the correct answer from the options given below : **[JEE (Main) 2024]**

(1) (A) – (IV), (B) – (III), (C) – (I), (D) – (II) (2) (A) – (III), (B) – (I), (C) – (II), (D) – (IV)

(3) (A) – (II), (B) – (IV), (C) – (III), (D) – (I) (4) (A) – (I), (B) – (II), (C) – (IV), (D) – (III)

CED335

18. Which among the following is incorrect statement? **[JEE (Main) 2024]**

(1) Electromeric effect dominates over inductive effect

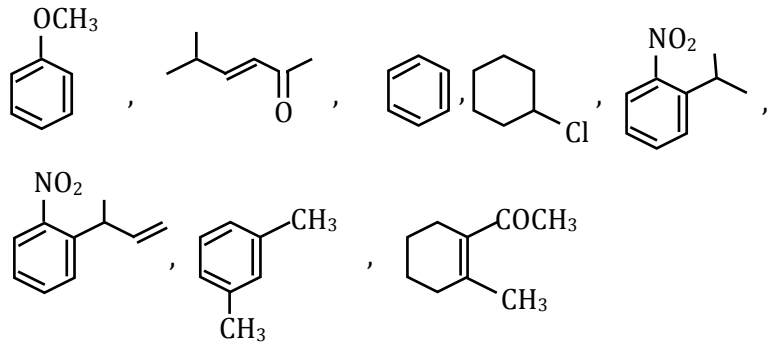
(2) The electromeric effect is, temporary effect

(3) The organic compound shows electromeric effect in the presence of the reagent only

(4) Hydrogen ion (H^+) shows negative electromeric effect

CED336

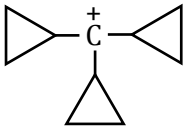
19. How many compounds among the following compounds show inductive, mesomeric as well as hyperconjugation effects? **[JEE (Main) 2024]**



CED337

For More Join: @IITJEE_Advanced

20. The decreasing order of hydride affinity for following carbocations is:

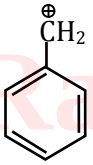
A.	$CH_2 = CH-\overset{+}{C}(CH_3)-CH_3$
B.	$C_6H_5-\overset{+}{C}(C_6H_5)-C_6H_5$
C.	$H_3C-\overset{+}{C}(CH_3)-CH_3$
D.	

Choose the correct answer from the options given below: **[JEE (Main) 2023]**

(1) A, C, B, D (2) C, A, B, D

(3) C, A, D, B (4) A, C, D, B

CED278

21.  $CH_2=\overset{\oplus}{C}H$ $CH_3-\overset{\oplus}{C}H_2$ $HC\equiv\overset{\oplus}{C}$

(A) (B) (C) (D)

[JEE (Main) 2021]

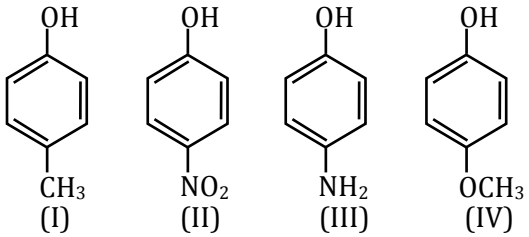
The correct order of stability of given carbocation is :

(1) A > C > B > D (2) D >B > C > A

(3) D > B > A > C (4) C > A > D > B

CED045

22. The increasing order of boiling points of the following compounds is : **[JEE (Main) 2021]**



(I) (II) (III) (IV)

(1) I < IV < III < II (2) IV < I < II < III (3) I < III < IV < II (4) III < I < II < IV

CED050

23. Which of the following has the shortest C-Cl bond? **[JEE (Main) 2021]**

(1) Cl–CH=CH–OCH_3 (2) Cl–CH=CH–CH_3

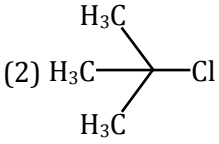
(3) Cl–CH=CH_2 (4) Cl–CH=CH–NO_2

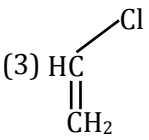
CED047

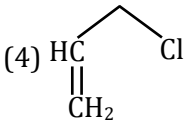
For More Join: @IITJEE_Advanced

24. Among the following compounds, which one has the shortest C—Cl bond ? **[JEE (Main) 2021]**

(1) $H_3C–Cl$

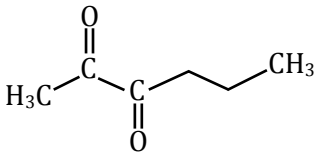
(2) 

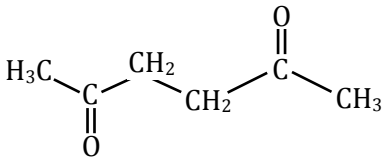
(3) 

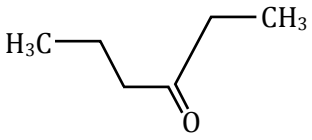
(4) 

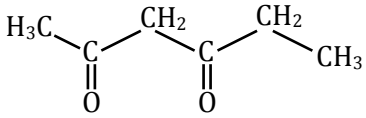
CED049

25. Which of the following has highly acidic hydrogen? **[JEE (Main) 2024]**

(1) 

(2) 

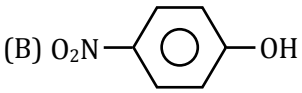
(3) 

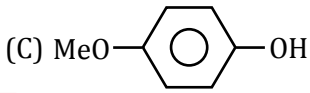
(4) 

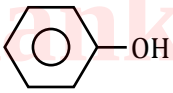
CED338

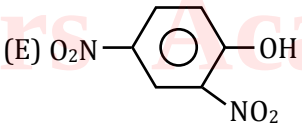
26. The ascending order of acidity of –OH group in the following compounds is :

(A) Bu – OH

(B) 

(C) 

(D) 

(E) 

Pankers Academy JEE

Choose the correct answer from the options given below : **[JEE (Main) 2024]**

(1) (A) < (D) < (C) < (B) < (E)

(2) (C) < (A) < (D) < (B) < (E)

(3) (C) < (D) < (B) < (A) < (E)

(4) (A) < (C) < (D) < (B) < (E)

CED339

27. Match List I with List II

List I (Compound)		**List II (pK_a value)**	
A.	Ethanol	I.	10.0
B.	Phenol	II.	15.9
C.	m-Nitrophenol	III.	7.1
D.	p-Nitrophenol	IV.	8.3

Choose the correct answer from the options given below :- **[JEE (Main) 2024]**

(1) A-I, B-II, C-III, D-IV

(2) A-IV, B-I, C-II, D-III

(3) A-III, B-IV, C-I, D-II

(4) A-II, B-I, C-IV, D-III

CED340

For More Join: @IITJEE_Advanced

28. The correct sequence of acidic strength of the following aliphatic acids in their decreasing order is :

CH_3CH_2COOH, CH_3COOH, $CH_3CH_2CH_2COOH$, HCOOH **[JEE (Main) 2024]**

(1) HCOOH > CH_3COOH > CH_3CH_2COOH > $CH_3CH_2CH_2COOH$

(2) HCOOH > $CH_3CH_2CH_2COOH$ > CH_3CH_2COOH > CH_3COOH

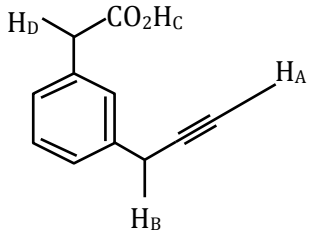
(3) $CH_3CH_2CH_2COOH$ > CH_3CH_2COOH > CH_3COOH > HCOOH

(4) CH_3COOH > CH_3CH_2COOH > $CH_3CH_2CH_2COOH$ > HCOOH

CED341

29. What is the correct order of acidity of the protons marked A–D in the given compounds?

[JEE (Main) 2023]



(1) $H_C > H_D > H_B > H_A$ (2) $H_C > H_D > H_A > H_B$

(3) $H_D > H_C > H_B > H_A$ (4) $H_C > H_A > H_D > H_B$

CED275

30. The descending order of acidity for the following carboxylic acid is :

A. CH_3COOH B. F_3C–COOH C. $ClCH_2$–COOH D. FCH_2–COOH

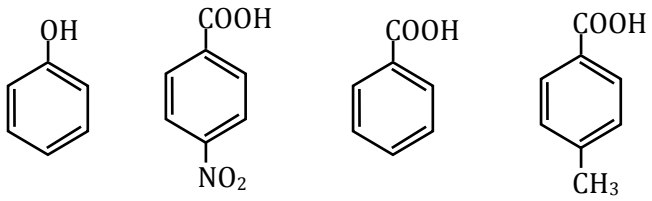
E. $BrCH_2$–COOH

Choose the correct answer from the options given below : **[JEE (Main) 2023]**

(1) D > B > A > E > C (2) E > D > B > A > C (3) B > C > D > E > A (4) B > D > C > E > A

CED277

31. The correct order of acid character of the following compounds is: **[JEE (Main) 2021]**

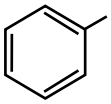


(I) (II) (III) (IV)

(1) III > II > I > IV (2) IV > III > II > I (3) I > II > III > IV (4) II > III > IV > I

CED032

32. Which among the following is the strongest acid ? **[JEE (Main) 2021]**

(1) $CH_3CH_2CH_2CH_3$ (2)  (3)  (4) 

CED043

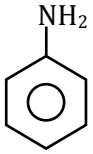
For More Join: @IITJEE_Advanced

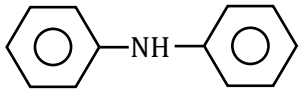
33. The correct order of nucleophilicity is **[JEE (Main) 2022]**

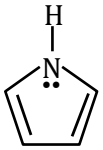
(1) $F^- > OH^-$ (2) $H_2\ddot{O} > OH^-$ (3) $R\ddot{O}H > RO^-$ (4) $NH_2^- > NH_3$

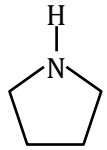
CED362

34. Which of the following is strongest Bronsted base? **[JEE (Main) 2024]**

(1) 

(2) 

(3) 

(4) 

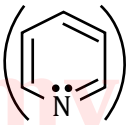
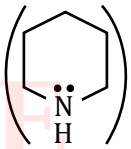
CED342

35. What will be the decreasing order of basic strength of the following conjugate bases ?
[JEE (Main) 2024]

(1) $C\bar{l} > OH > R\bar{O} > CH_3C\bar{O}O$ (2) $R\bar{O} > {}^-OH > CH_3C\bar{O}O > C\bar{l}$

(3) ${}^-OH > R\bar{O} > CH_3C\bar{O}O > C\bar{l}$ (4) $C\bar{l} > R\bar{O} > {}^-OH > CH_3C\bar{O}O$

CED343

36. Correct order of basic strength of Pyrrole  , Pyridine  and Piperidine 
is: **[JEE (Main) 2024]**

(1) Piperidine > Pyridine > Pyrrole (2) Pyrrole > Pyridine > Piperidine

(3) Pyridine > Piperidine > Pyrrole (4) Pyrrole > Piperidine >Pyridine

CED344

37. A. Phenyl methanamine **[JEE (Main) 2021]**
B. N,N-Dimethylaniline
C. N-Methyl aniline
D. Benzenamine
Choose the correct order of basic nature of the above amines.
(1) A > C > B > D (2) D > C > B > A (3) D > B > C > A (4) A > B > C > D

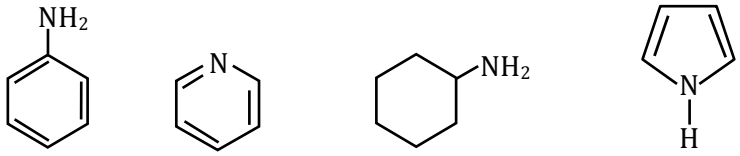
CED033

38. Given below are two statements : **[JEE (Main) 2021]**
Statement I : Aniline is less basic than acetamide.
Statement II : In aniline, the lone pair of electrons on nitrogen atom is delocalised over benzene ring due to resonance and hence less available to a proton.
Choose the **most appropriate** option ;
(1) Statement I is true but statement II is false.
(2) Statement I is false but statement II is true.
(3) Both statement I and statement II are true.
(4) Both statement I and statement II are false.

CED044

Rankers Academy JEE
For More Join: @IITJEE_Advanced

39. The decreasing order of basicity of the following amines is : **[JEE (Main) 2021]**

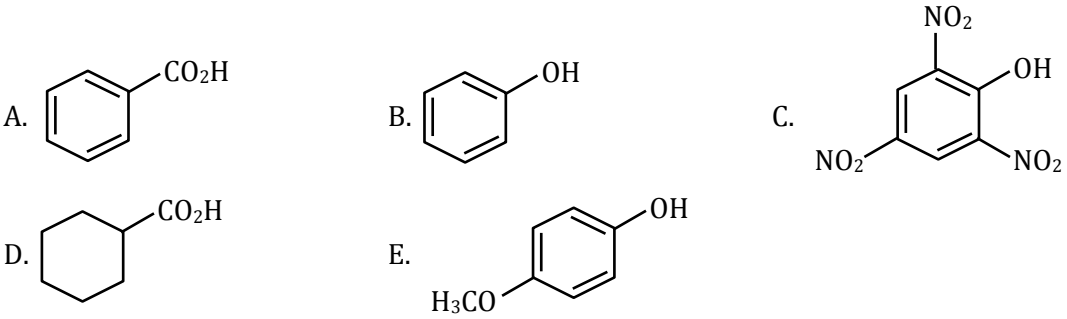


(I) (II) (III) (IV)

(1) (I) > (III) > (IV) > (II)
(2) (III) > (I) > (II) > (IV)
(3) (III) > (II) > (I) > (IV)
(4) (II) > (III) > (IV) > (I)

CED048

40. The compounds that produce CO_2 with aqueous $NaHCO_3$ solution are:



Choose the ***correct*** answer from the options given below : **[JEE (Main) 2025]**

(1) A and C only
(2) A, B and E only
(3) A, C and D only
(4) A and B only

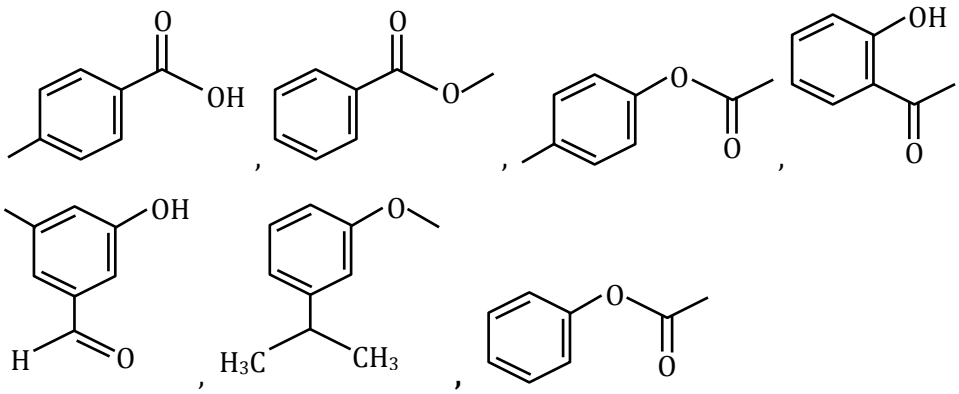
CED376

41. Which one the following compounds will readily react with dilute NaOH? **[JEE (Main) 2024]**

(1) $C_6H_5CH_2OH$
(2) C_2H_5OH
(3) $(CH_3)_3COH$
(4) C_6H_5OH

CED345

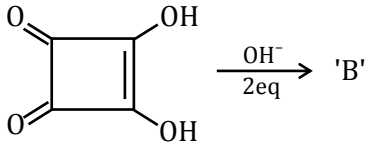
42. Number of compounds from the following which will not dissolve in cold $NaHCO_3$ and NaOH solutions but will dissolve in hot NaOH solution is ____ . **[JEE (Main) 2023]**



CED276

For More Join: @IITJEE_Advanced

43. Correct statements for the given reaction are: **[JEE (Main) 2023]**

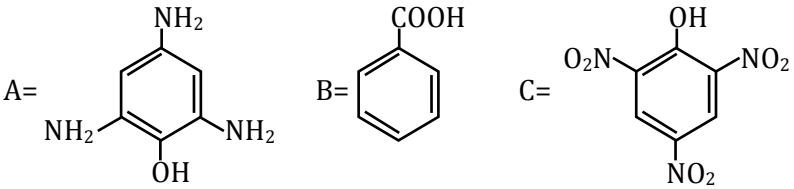


'A'

A. Compound 'B' is aromatic
B. The completion of above reaction is very slow
C. 'A' shows tautomerism
D. The bond lengths C-C in compound B are found to be same
Choose the correct answer from the options given below :
(1) A, B and D only
(2) A, B and C only
(3) B, C and D only
(4) A, C and D only

CED279

44. Compound(s) which will liberate carbon dioxide with sodium bicarbonate solution is/are: **[JEE (Main) 2021]**



(1) B only
(2) C only
(3) B and C only
(4) A and B only

CED034

45. The purification method based on the following physical transformation is : **[JEE (Main) 2025]**

$$\underset{(X)}{\text{Solid}} \xrightarrow{\text{Heat}} \underset{(X)}{\text{Vapour}} \xrightarrow{\text{Cool}} \underset{(X)}{\text{Solid}}$$

(1) Sublimation
(2) Distillation
(3) Crystallization
(4) Extraction

CED377

46. The technique used for purification of steam volatile water immiscible substance is: **[JEE (Main) 2024]**

(1) Fractional distillation
(2) Fractional distillation under reduced pressure
(3) Distillation
(4) Steam distillation

CED346

47. Match List-I with List-II

LIST-I (Technique)		LIST-II (Application)	
A.	Distillation	I.	Separation of glycerol from spent-lye
B.	Fractional distillation	II.	Aniline - Water mixture
C.	Steam distillation	III.	Separation of crude oil fractions
D.	Distillation under reduced pressure	IV.	Chloroform-Aniline

Choose the correct answer from the options given below: **[JEE (Main) 2024]**
(1) A-IV, B-I, C-II, D-III
(2) A-IV, B-III, C-II. D-I
(3) A-I. B-II, C-IV, D-III
(4) A-II, B-III. C-I, D-IV

CED347

For More Join: @IITJEE_Advanced

48. The fragrance of flowers is due to the presence of some steam volatile organic compounds called essential oils. These are generally insoluble in water at room temperature but are miscible with water vapour in vapour phase. A suitable method for the extraction of these oils from the flowers is -

[JEE (Main) 2024]

(1) Crystallisation
(2) Distillation under reduced pressure
(3) Distillation
(4) Steam distillation

CED348

49. Which of the following statements are correct?

A. Glycerol is purified by vacuum distillation because it decomposes at its normal boiling point.
B. Aniline can be purified by steam distillation as aniline is miscible in water.
C. Ethanol can be separated from ethanol water mixture by azeotropic distillation because it forms azeotrope.
D. An organic compound is pure, if mixed M.P. is remained same.

Choose the **most appropriate** answer from the options given below : **[JEE (Main) 2024]**

(1) A, B, C only
(2) A, C, D only
(3) B, C, D only
(4) A, B, D only

CED349

50. Given below are two statements :

Statement (I) : In partition chromatography, stationary phase is thin film of liquid present in the inert support.

Statement (II) : In paper chromatography, the material of paper acts as a stationary phase.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE (Main) 2025]

(1) Both **Statement I** and **Statement II** are false
(2) **Statement I** is true but **Statement II** is false
(3) Both **Statement I** and **Statement II** are true
(4) **Statement I** is false but **Statement II** is true

CED378

51. Chromatographic technique/s based on the principle of differential adsorption is/are

A. Column chromatography
B. Thin layer chromatography
C. Paper chromatography

Choose the most appropriate answer from the options given below: **[JEE (Main) 2024]**

(1) B only
(2) A only
(3) A & B only
(4) C only

CED350

52. Which among the following purification methods is based on the principle of "Solubility" in two different solvents? **[JEE (Main) 2024]**

(1) Column Chromatography
(2) Sublimation
(3) Distillation
(4) Differential Extraction

CED351

For More Join: @IITJEE_Advanced

53. The adsorbent used in adsorption chromatography is/are

A. silica gel B. alumina C. quick lime D. magnesia

Choose the **most appropriate** answer from the options given below : **[JEE (Main) 2024]**

(1) B only (2) C and D only (3) A and B only (4) A only

CED352

54. The correct statements among the following, for a "chromatography" purification method is: **[JEE (Main) 2024]**

(1) Organic compounds run faster than solvent in the thin layer chromatographic plate.

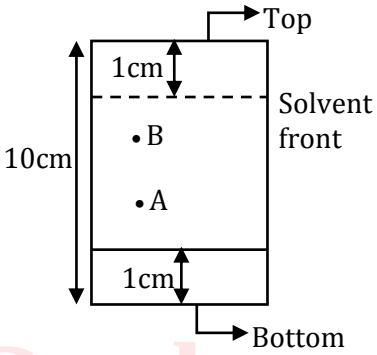
(2) Non-polar compounds are retained at top and polar compounds come down in column chromatography.

(3) R_f of a polar compound is smaller than that of a non-polar compound.

(4) R_f is an integral value.

CED353

55.



In the given TLC, the distance of spot A & B are 5 cm & 7 cm, from the bottom of TLC plate, respectively.

R_f value of B is $x \times 10^{-1}$ times more than A. The value of x is____. **[JEE (Main) 2024]**

CED354

56. Using the given figure, the ratio of R_f values of sample A and sample C is $x \times 10^{-2}$. Value of x is _____. **[JEE (Main) 2024]**

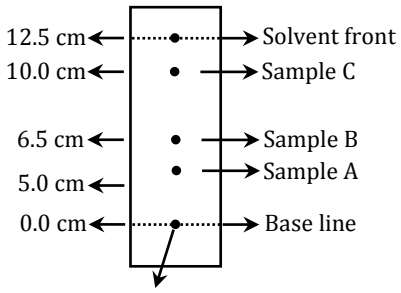


Fig : Paper chromatography of Samples

CED355

57. Methods used for purification of organic compounds are based on : **[JEE (Main) 2024]**

(1) neither on nature of compound nor on the impurity present.

(2) nature of compound only.

(3) nature of compound and presence of impurity.

(4) presence of impurity only.

CED356

For More Join: @IITJEE_Advanced

58. Given below are two statement :

Statement (I) : Nitrogen, sulphur, halogen and phosphorus present in an organic compound are detected by Lassaigne's Test.

Statement (II) : The elements present in the compound are converted from covalent form into ionic form by fusing the compound with Magnesium in Lassaigne's test.

In the light of the above statements, choose the **correct** answer from the options given below :

[JEE (Main) 2025]

(1) Both Statement I and Statement II are true

(2) Both Statement I and Statement II are false

(3) Statement I is true but Statement II is false

(4) Statement I is false but Statement II is true

CED379

59. Appearance of blood red colour, on treatment of the sodium fusion extract of an organic compound with $FeSO_4$ in presence of concentrated H_2SO_4 indicates the presence of element/s

[JEE (Main) 2024]

(1) Br (2) N (3) N and S (4) S

CED357

60. Lassaigne's test is used for detection of : **[JEE (Main) 2024]**

(1) Nitrogen and Sulphur only

(2) Nitrogen, Sulphur and Phosphorous Only

(3) Phosphorous and halogens only

(4) Nitrogen, Sulphur, phosphorous and halogens

CED358

Rankers Academy JEE

61. In the sulphur estimation, 0.20 g of a pure organic compound gave 0.40 g of barium sulphate. The percentage of sulphur in the compound is ___ $\times 10^{-1}$ %.

(Molar mass : O = 16, S = 32, Ba = 137 in g mol^{-1})

[JEE (Main) 2025]

CED380

62. In Carius method of estimation of halogen, 0.25 g of an organic compound gave 0.15 g of silver bromide (AgBr). The percentage of Bromine in the organic compound is _____ $\times 10^{-1}$% (Nearest integer).

(Given : Molar mass of Ag is 108 and Br is 80 g mol^{-1})

[JEE (Main) 2025]

CED381

63. Given below are two statements I and II.

Statement I : Dumas method is used for estimation of "Nitrogen" in an organic compound.

Statement II : Dumas method involves the formation of ammonium sulphate by heating the organic compound with conc H_2SO_4.

In the light of the above statements, choose the ***correct*** answer from the options given below

[JEE (Main) 2025]

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are false

(4) Statement I is true but Statement II is false

CED382

For More Join: @IITJEE_Advanced

64. An organic compound has 42.1% carbon, 6.4% hydrogen and remainder is oxygen. If its molecular weight is 342, then its molecular formula is :

[JEE (Main) 2024]

(1) $C_{11}H_{18}O_{12}$ (2) $C_{12}H_{20}O_{12}$

(3) $C_{14}H_{20}O_{10}$ (4) $C_{12}H_{22}O_{11}$

CED359

65. Given below are two statements:

Statement (I) : Kjeldahl method is applicable to estimate nitrogen in pyridine.

Statement (II) : The nitrogen present in pyridine can easily be converted into ammonium sulphate in Kjeldahl method.

In the light of the above statements, choose the **correct** answer from the options given below.

[JEE (Main) 2024]

(1) Both **Statement I** and **Statement II** is false

(2) **Statement I** is false but **Statement II** is true

(3) Both **Statement I** and **Statement II** is true

(4) **Statement I** is true but **Statement II** is false

CED360

66. In Kjeldahl's method for estimation of nitrogen, $CuSO_4$ acts as :

[JEE (Main) 2024]

(1) Reducing agent (2) Catalytic agent

(3) Hydrolysis agent (4) Oxidising agent

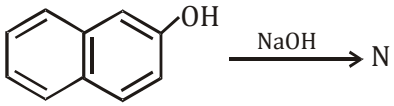
CED361

Rankers Academy JEE

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. The number of resonance structures for N is: **[JEE (Advanced) 2015]**



CED082

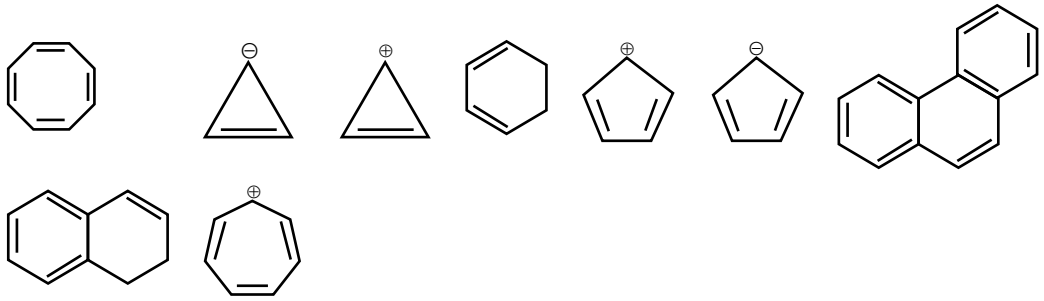
2. The correct stability order of the following resonance structures is **[JEE (Advanced) 2009]**

(I) $H_2C=\overset{\oplus}{N}=\overset{\ominus}{N}$ (II) $H_2\overset{\oplus}{C}-N=\overset{\ominus}{N}$ (III) $H_2\overset{\ominus}{C}-\overset{\oplus}{N}=N$ (IV) $H_2\overset{\ominus}{C}-N=\overset{\oplus}{N}$

(A) (I) > (II) > (IV) > (III)
(B) (I) > (III) > (II) > (IV)
(C) (II) > (I) > (III) > (IV)
(D) (III) > (I) > (IV) > (II)

CED071

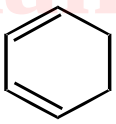
3. Among the following, the number of aromatic compound (s) is- **[JEE (Advanced) 2017]**

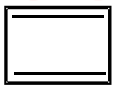


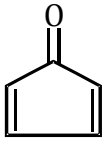
CED084

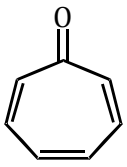
4. Which of the following molecules, in pure from, is (are) unstable at room temperature? **[JEE (Advanced) 2012]**

Rankers Academy JEE

(A) 

(B) 

(C) 

(D) 

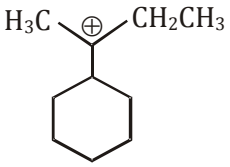
CED077

5. The hyper conjugative stabilities of tert-butyl cation and 2-butene, respectively, are due to **[JEE (Advanced) 2013]**

(A) $\sigma \rightarrow$ p (empty) and $\sigma \rightarrow \pi^*$ electron delocalisations
(B) $\sigma \rightarrow \sigma^*$ and $\sigma \rightarrow \pi$ electron delocalisations
(C) $\sigma \rightarrow$ p (filled) and $\sigma \rightarrow \pi$ electron delocalisations
(D) p (filled) $\rightarrow \sigma^*$ and $\sigma \rightarrow \pi^*$ electron delocalisations

CED079

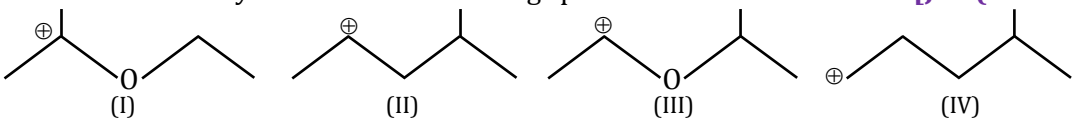
6. The total number of contributing structures showing hyperconjugation (involving C–H bonds) for the following carbocation is. **[JEE (Advanced) 2011]**



CED074

For More Join: @IITJEE_Advanced

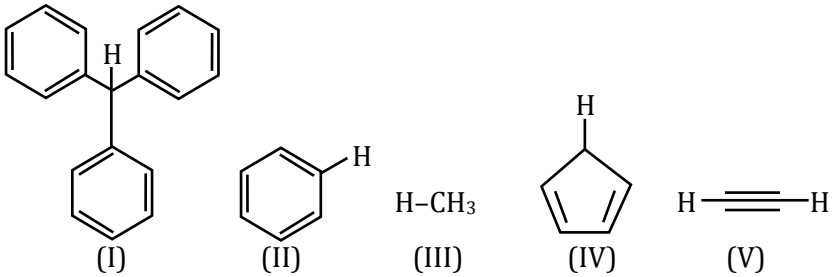
7. The correct stability order for the following species is : **[JEE (Advanced) 2008]**



(A) II > IV > I > III
(B) I > II > III > IV
(C) II > I > IV > III
(D) I > III > II > IV

CED069

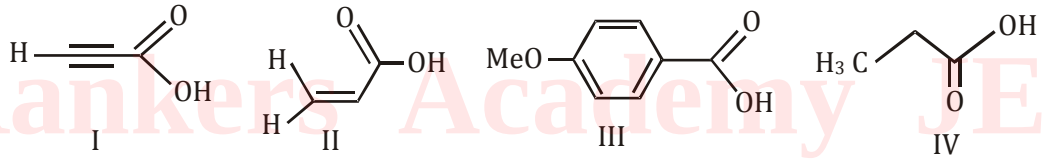
8. With respect to the compounds **I-V**, choose the correct statement(s). **[JEE (Advanced) 2020]**



(A) The acidity of compound **I** is due to delocalization in the conjugate base.
(B) The conjugate base of compound **IV** is aromatic.
(C) Compound **II** becomes more acidic, when it has a -NO_2 substituent.
(D) The acidity of compounds follows the order **I > IV > V > II > III.**

CED086

9. The correct order of acid strength of the following carboxylic acids is – **[JEE (Advanced) 2019]**



Rankers Academy JEE

(A) I > III > II > IV
(B) III > II > I > IV
(C) II > I > IV > III
(D) I > II > III > IV

CED085

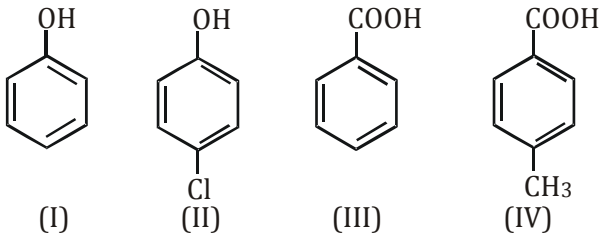
10. The carboxyl functional group (–COOH) is present in – **[JEE (Advanced) 2012]**
(A) picric acid
(B) barbituric acid
(C) ascorbic acid
(D) aspirin

CED075

11. Among the following compounds, the most acidic is **[JEE (Advanced) 2011]**
(A) p-nitrophenol
(B) p-hydroxybenzoic acid
(C) o-hydroxybenzoic acid
(D) p-toluic acid

CED073

12. The correct acidity order of the following is: **[JEE (Advanced) 2009]**

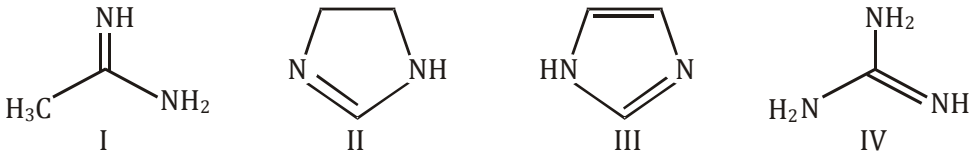


(A) (III) > (IV) > (II) > (I)
(B) (IV) > (III) > (I) > (II)
(C) (III) > (II) > (I) > (IV)
(D) (II) > (III) > (IV) > (I)

CED070

For More Join: @IITJEE_Advanced

13. The order of basicity among the following compounds is **[JEE (Advanced) 2017]**



(A) II > I > IV > III
(B) IV > II > III > I
(C) I > IV > III > II
(D) IV > I > II > III

CED083

14. Hydrogen bonding plays a central role in the following phenomena **[JEE (Advanced) 2014]**
(A) Ice floats in water
(B) Higher Lewis basicity of primary amines than tertiary amines in aqueous solutions
(C) Formic acid is more acidic than acetic acid
(D) Dimerisation of acetic acid in benzene

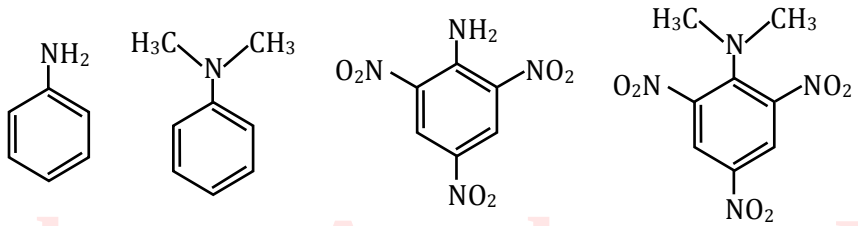
CED081

15. The total number of lone-pairs of electrons in melamine is **[JEE (Advanced) 2013]**

CED080

16. Consider the following four compounds **I**, **II**, **III**, and **IV**.

[JEE (Advanced) 2020]



(I) (II) (III) (IV)

Pankers Academy JEE

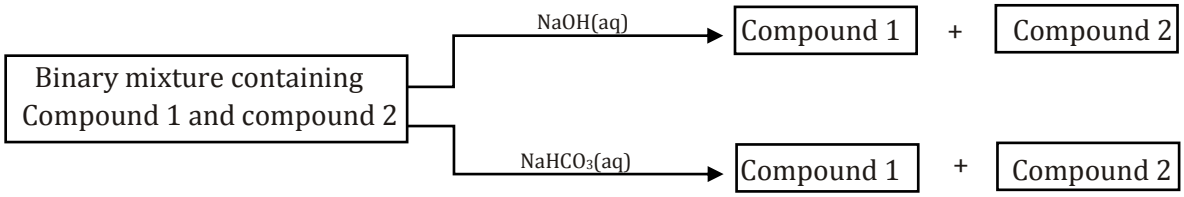
Choose the correct statement(s).
(A) The order of basicity is **II** > **I** > **III** > **IV**.
(B) The magnitude of pK_b difference between **I** and **II** is more than that between **III** and **IV**.
(C) Resonance effect is more in **III** than in **IV**.
(D) Steric effect makes compound **IV** more basic than **III**.

CED088

17. The compound that does NOT liberate CO_2, on treatment with aqueous sodium bicarbonate solution, is – **[JEE (Advanced) 2013]**
(A) Benzoic acid
(B) Benzenesulphonic acid
(C) Salicylic acid
(D) Carbolic acid (phenol)

CED078

18. Identify the binary mixtures (s) that can be separated into the individual compounds, by differential extraction, as shown in the given scheme – **[JEE (Advanced) 2012]**



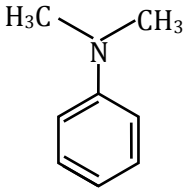
(A) C_6H_5OH and C_6H_5 COOH
(B) C_6H_5COOH and $C_6H_5CH_2OH$
(C) $C_6H_5CH_2OH$ and C_6H_5OH
(D) $C_6H_5CH_2OH$ and $C_6H_5CH_2COOH$

CED076

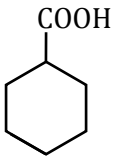
For More Join: @IITJEE_Advanced

19. Amongst the following, the total number of compounds soluble in aqueous NaOH is:

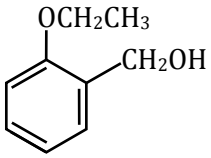
[JEE (Advanced) 2010]



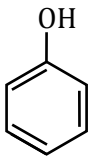
(A)



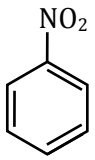
(B)



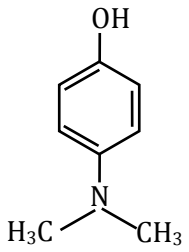
(C)



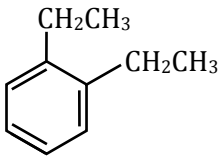
(D)



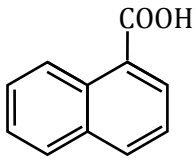
(E)



(F)



(G)



(H)

CED072

20. Match the compounds in LIST-I with the observation in LIST-II, and choose the correct option.

	LIST-I		**LIST-II**
(I)	Aniline	(P)	Sodium fusion extract of the compound on boiling with $FeSO_4$, followed by acidification with conc. H_2SO_4, gives Prussian blue color.
(II)	o-Cresol	(Q)	Sodium fusion extract of the compound on treatment with sodium nitroprusside gives blood red color.
(III)	Cysteine	(R)	Addition of the compound to a saturated solution of $NaHCO_3$ results in effervescence.
(IV)	Caprolactam	(S)	The compound reacts with bromine water to give a white precipitate.
		(T)	Treating the compound with neutral $FeCl_3$ solution produces violet colour.

[JEE (Advanced) 2022]

(A) I → P, Q; II → S; III → Q, R; IV → P
(B) I → P ; II → R, S; III → R; IV → Q, S
(C) I → Q, S; II → P, T; III → P; IV → S
(D) I → P, S; II → T; III → Q, R; IV → P

CED363

Rankers Academy JEE

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	D	D	A	A	D	A	A	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	B	D	D	B	A	D	A	A	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	C	C	A	D	C	B	D	D	B	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	D	D	C	C	A	B	B	D	B	C
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	B	B	D	B	D	B	C	A	D	B
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	C	C	C	D	A	D	A	D	B	C
Que.	61	62	63	64	65	66	67	68	69	70
Ans.	D	A	B	A	D	A	C	D	C	B
Que.	71	72	73	74	75	76	77	78	79	80
Ans.	B	C	C	C	D	B	A	B	D	D
Que.	81	82	83	84	85	86	87	88	89	90
Ans.	A	C	D	C	C	D	C	B	A	C
Que.	91	92	93	94	95	96	97	98	99	100
Ans.	A	A	A	C	C	A	C	A	D	A
Que.	101	102	103	104	105	106	107	108	109	110
Ans.	B	C	A	D	B	D	C	A	B	B
Que.	111	112	113	114	115	116	117	118	119	120
Ans.	B	D	B	D	D	C	C	A	C	B

Bankers Academy JEE

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	CD	CD	AB	BC	BD	AD	ABC	ABC	BD	C
Que.	11	12	13	14						
Ans.	B	D	A	C						

EXERCISE (S)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	5	3	8	3	3	4	2	4	6
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	3	4	6	3	2	6	5	5	3
Que.	21	22	23	24	25	26	27			
Ans.	6	2	6	56	70	28	51.2			

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	3	3	1	1	4	3	1	1	1	2
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	5	3	4	2	1	3	1	4	4	2
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	1	1	4	3	4	4	4	1	2	4
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	4	4	4	4	2	1	4	2	3	3
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	4	3	4	3	1	4	2	4	2	2
Que.	51	52	53	54	55	56	57	58	59	60
Ans.	3	4	3	3	15	50	3	3	3	4
Que.	61	62	63	64	65	66				
Ans.	275	255	4	4	1	2				

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	9	B	5	BC	A	6	D	ABC	D	D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	D	ABD	6	CD	D	BD	4	D

Rankers Academy JEE

For More Join: @IITJEE_Advanced